Linear-in-n (1+epsilon)-approximation for the general planar two-line-center problem

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Editorial scope. Original native socg_full.tex214–2894 (body1–2681), complete sections1–8 and all appendices. Count general-main theorem only; retain other orientation variants and supporting certificate/constant approximation for full context without separate counts. The algorithm constructs a constant-size anchor-pair candidate set, builds a linear-time constant approximation and an O(1/epsilon^2) certificate, and reduces the general problem to O(1/epsilon) candidate slab orientations. Its fixed-orientation subroutine maintains constrained convex-hull widths under sliding-window updates, with amortized tangent/antipodal searches and exact optimization. All original appendix proofs, including the 10-approximation, optimization, lower-bound and zero-width cases, are included. The linked algorithm suite and its certificate/support results are counted together.

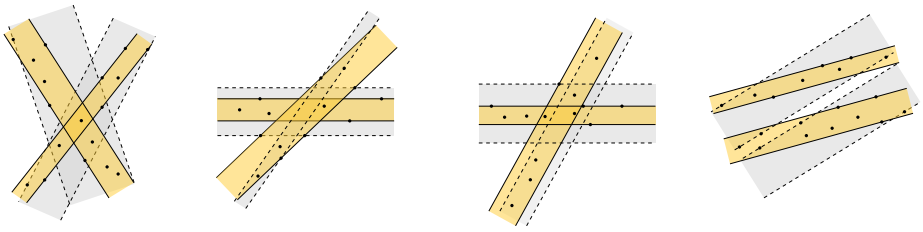
Allowed standard prerequisites (editorial).

- Author-cited standard planar convex hull/width results, Chan online constant width approximation, and dynamic convex-hull updates.. All exact native theorem statements/citations and original paper.bib are preserved. Original source-specific anchor/certificate, data-structure maintenance and approximation proofs, including Appendices A–D, are fully retained.

1 Introduction

Given a set S of n points in the plane, we study the two-line-center problem: finding two lines that minimize the maximum distance from each point in S to its closest line. This is a special case of the k-line-center problem, which asks, for a given set S of n points in the plane, to find k lines minimizing the maximum distance from each point in S to its closest line. A more general framework is projective clustering: Given $P \subset \mathbb{R}^d$, the goal is to find k flats of dimension j that best fit P under a given distance measure. It has wide applications in data mining [4, 5], unsupervised learning [27], database management [11], and computer vision [28].

The k-line-center problem is known to be NP-hard [25] when k is part of the input (even in the plane), and various approaches including approximation algorithms and randomized methods [1, 3, 7, 9, 12, 14, 18, 19, 21] have been studied.



(a) General (b) One fixed orientation (c) Two fixed orientations (d) Parallel

Figure 1: In each case, both yellow and gray pairs satisfy the restriction, but the yellow pair has the smaller maximum width.

When $k = 1$, the problem becomes computing the width of a point set. This can be done exactly in $O(n \log n)$ time [26], and an $\Omega(n \log n)$ lower bound is known [23]. A $(1+\varepsilon)$-approximation algorithm running in $O(n+1/\varepsilon)$ time was given in [12].

When $k = 2$, i.e., the two-line-center problem, an exact algorithm with running time $O(n^2 \log^2 n)$ is known [22]. The authors in [2] presented a $(1 + \varepsilon)$-approximation algorithm that runs in $O(n \log n + (n/\varepsilon^2) \log(1/\varepsilon) + \varepsilon^{-7/2} \log(1/\varepsilon))$ time. Later, as the $(1 + \varepsilon)$-approximation algorithm for the width problem was improved from $O(n + \varepsilon^{-3/2})$ to $O(n + 1/\varepsilon)$ [12], their algorithm is consequently improved to $O(n \log n + (n/\varepsilon^2) \log(1/\varepsilon) + (1/\varepsilon^3) \log(1/\varepsilon))$, even though it is not explicitly mentioned.

Observe that the two-line-center problem is equivalent to finding a pair of slabs whose union covers the input point set while minimizing their maximum width.

We study the two-line-center problem and three natural variants. Let $\mu = \min\{n, 1/\varepsilon^2\}$, and let w^* denote the maximum width of an optimal pair of slabs in each problem. Each variant is illustrated in Figure 1. Table 1 summarizes the previous and new results.

1. **General:** Given a set S of n points in the plane, we study the problem of finding a pair of slabs whose union covers S while minimizing the maximum of their widths. We present an $O(n + (\mu/\varepsilon) \log \mu)$-time algorithm that finds a pair of slabs whose union covers S and whose maximum width is at most $(1 + \varepsilon)w^*$. The running time can be expressed as both $O((n/\varepsilon) \log(1/\varepsilon))$ and $O(n + (1/\varepsilon^3) \log(1/\varepsilon))$.

2. **One Fixed Orientation:** Given a set S of n points in the plane and an orientation θ, we study the problem of finding a pair of slabs such that one of them has orientation θ and their union covers S while minimizing the maximum of their widths. We present an exact algorithm that runs in $O(n \log n)$ time, improving the previously best $O(n \log^3 n)$-time algorithm [6], and show that this is optimal.

   We also present an $O(n + \mu \log \mu)$-time algorithm that finds a pair of slabs such that one of them has orientation θ, their union covers S, and their maximum width is at most $(1 + \varepsilon)w^*$. The running time can be expressed as $O(n \log(1/\varepsilon))$ and $O(n + 1/\varepsilon^2 \log(1/\varepsilon))$.

	Exact	**Approximation**
General	$O(n^2 \log^2 n)$ [22]	$O(n \log n + \frac{n}{\varepsilon^2} \log \frac{1}{\varepsilon} + \frac{1}{\varepsilon^3} \log \frac{1}{\varepsilon})$ [2, 12]
		$\boldsymbol{O(\frac{n}{\varepsilon} \log \frac{1}{\varepsilon})}$, $\boldsymbol{O(n + \frac{1}{\varepsilon^3} \log \frac{1}{\varepsilon})}$ (Sec. 5)
One fixed orientation	$O(n \log^3 n)$ [6]	-
	$\boldsymbol{O(n \log n)}$ (Sec. 4)	$\boldsymbol{O(n \log \frac{1}{\varepsilon})}$, $\boldsymbol{O(n + \frac{1}{\varepsilon^2} \log \frac{1}{\varepsilon})}$ (Sec. 4)
Two fixed orientations	$O(n \log n)$ [6]	-
		$\boldsymbol{O(n + \frac{1}{\varepsilon})}$ (Sec. 6)
Parallel	$O(n^2)$ [8]	-
		$\boldsymbol{O(\frac{n}{\varepsilon})}$, $\boldsymbol{O(n + \frac{1}{\varepsilon^3})}$ (Sec. 7)

Table 1: A summary of our results and previous work.

3. **Two Fixed Orientations:** Given a set S of n points in the plane and two orientations, θ_1 and θ_2, we study the problem of finding two slabs, one has orientation θ_1 and the other has orientation θ_2, whose union covers S while minimizing the maximum of their widths. We present an $O(n+1/\varepsilon)$-time algorithm that finds a pair of slabs such that one has orientation θ_1 and the other has orientation θ_2, their union covers S, and their maximum width is at most $(1+\varepsilon)w^*$.

4. **Parallel:** Given a set S of n points in the plane, we study the problem of finding a pair of parallel slabs whose union covers S while minimizing the maximum of their widths. We present an $O(n+\mu/\varepsilon)$-time algorithm that finds a pair of parallel slabs such that their union covers S and their maximum width is at most $(1+\varepsilon)w^*$.

1.1 Outline

In Section 2, we introduce the notation that will be used throughout the paper.

In Section 3, we introduce three key ingredients used throughout the paper. In Section 3.1, we modify the concept of an *anchor pair* (as introduced in [2]) and show how to compute it. In Section 3.2, we use an anchor pair to compute a 10-approximation for the general two-line-center problem in $O(n)$ time. In Section 3.3, we recall the concept of an *ε-certificate* (as defined in [3]), a subset of the input point set that suffices to obtain a $(1+\varepsilon)$-approximate solution, and show how to compute it in $O(n)$ time using an anchor pair from Section 3.1 and a 10-approximate solution from Section 3.2.

In Section 4, we address the one-fixed-orientation two-line-center problem. In Section 4.2, we give an $O(n)$-time decision algorithm, assuming the points are sorted. In Section 4.3, we present an $O(n \log n)$-time exact algorithm and prove its optimality in Appendix C. In Section 4.4, we give a $(1+\varepsilon)$-approximation algorithm.

In Section 5, we address the general two-line-center problem. Using an anchor pair (Section 3.1) and a 10-approximate solution (Section 3.2), we compute a candidate set of orientations for one of the slabs of a $(1+\varepsilon)$-approximate solution. Combining this with the algorithm from Section 4 and an ε-certificate (Section 3.3), we obtain a $(1+\varepsilon)$-approximation algorithm.

In Section 6, we address the two-fixed-orientations two-line-center problem. We first give a 2-approximation algorithm in Section 6.1. Based on this, we present a $(1+\varepsilon)$-approximation algorithm in Section 6.2.

In Section 7, we address the parallel two-line-center problem. We use the *gap-ratio* of a pair of slabs (as defined in [13]) and handle two cases separately: when the gap-ratio of an optimal pair is small (Section 7.1) and when it is large (Section 7.2). Combining these with an ε-certificate (Section 3.3), we present a $(1+\varepsilon)$-approximation algorithm.

2 Preliminaries

For a set P of points in the plane, we denote its diameter by $\mathrm{diam}(P)$, its width by $\mathrm{width}(P)$, and its convex hull by $\mathrm{conv}(P)$. If $P = \emptyset$, then $\mathrm{width}(P) = 0$. For a slab σ, we denote its width by $w(\sigma)$, which is the distance between the two bounding lines of σ (see Figure 2(a)).

For a line ℓ, we say that ℓ has orientation $\theta \in [0, \pi)$ if θ is the counterclockwise angle from the x-axis to ℓ. A slab σ is said to have orientation $\theta \in [0, \pi)$ if its two bounding lines have orientation θ (see Figure 2(a)). For a finite set P of points in the plane, let $\sigma_\theta(P)$ denote the minimum-width slab of orientation θ that encloses P, and define $\mathrm{width}_\theta(P) = w(\sigma_\theta(P))$ (see Figure 2(b)).

For a set P of points in the plane and an orientation $\theta \in [0, \pi)$, we say that a pair (p, q) with $p, q \in P$ is an *antipodal pair* of P with respect to θ if p lies on one bounding line of $\sigma_\theta(P)$ and q lies on the other (see Figure 2(b)).

For two finite sets A and B in the plane, let $d(A, B)$ denote the minimum distance between a point of A and a point of B. For two points p and q in the plane, let ℓ_{pq} be the line passing through p and q.

For each of the four problems, we assume that $w^* > 0$, where w^* denotes the maximum width of an optimal pair of slabs for that problem. The case where

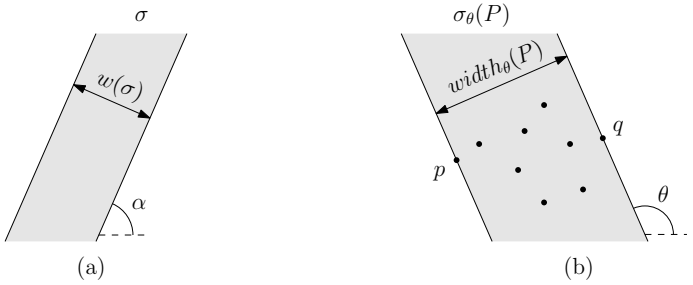


Figure 2: (a) The orientation of σ is α. (b) For a set P of points and an orientation θ, $\sigma_\theta(P)$ and $\mathrm{width}_\theta(P)$ are shown. Observe that (p, q) is an antipodal pair of P with respect to θ.

$w^* = 0$ can be detected and handled in linear time. See Appendix D.

Throughout the paper, let $\varepsilon > 0$ be a sufficiently small real number and let S denote an input set of n points in the plane.

3 Ingredients

In this section, we describe three key ingredients used throughout the paper: (1) an anchor pair, (2) a linear-time 10-approximation algorithm, and (3) an ε-certificate.

Throughout this section, we denote by $\Sigma^* = (\sigma_1^*, \sigma_2^*)$ an optimal pair of slabs for the general two-line-center problem, and let $w^* = \max\{w(\sigma_1^*), w(\sigma_2^*)\}$.

3.1 An Anchor Pair and Its Candidate Set

We first define an anchor pair and show that a candidate set for an anchor pair of Σ^* of constant size can be computed in $O(n)$ time. Recall that S denotes the input set of n points in the plane.

Definition 1 (Anchor pair)**.** *For a pair of slabs $\Sigma = (\sigma_1, \sigma_2)$, we call a pair of points $p, q \in \mathbb{R}^2$ an **anchor pair of** Σ if*

(1) $p, q \in \sigma_1$ and $d(p, q) \geq \mathrm{diam}((\sigma_1 \setminus \sigma_2) \cap S)/4$, or

(2) $p, q \in \sigma_2$ and $d(p, q) \geq \mathrm{diam}((\sigma_2 \setminus \sigma_1) \cap S)/4$.

Note that p or q may lie in $\sigma_1 \cap \sigma_2$. (See Figure 3.)

We remark that the authors in [2] used the concept of an anchor pair, defined slightly differently. They showed that a candidate set of constant size can be computed in $O(n \log n)$ time. In the proof of the following lemma, we point out a gap in their proof. By adopting our definition of an anchor pair, we both resolve this gap and obtain its candidate set in $O(n)$ time.

Lemma 2. *We can compute, in $O(n)$ time, a set of at most 11 pairs of points such that for any pair Σ of slabs whose union covers S, at least one of them is an anchor pair of Σ.*

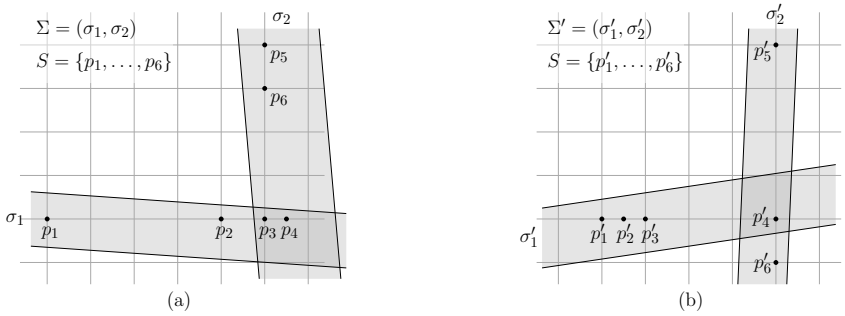


Figure 3: (a) Both (p_2, p_3) and (p_3, p_4) are anchor pairs of Σ, satisfying **(1)** and **(2)**, respectively. (b) (p'_1, p'_2) is an anchor pair of Σ', satisfying **(1)**, but (p'_4, p'_6) is not.

Proof. Let $\Sigma = (\sigma_1, \sigma_2)$ be a pair of slabs whose union covers S. Let Δ denote the diameter of S. Let p and q be a pair of points in S satisfying $d(p, q) \geq \Delta/2$, which can be found in $O(n)$ time. Let $R = d(p, q)/2$. We consider two open disks, D_p and D_q, of radius R centered at p and q, respectively. That is $D_p \cap D_q = \emptyset$.

If there exists $r \in S \setminus (D_p \cup D_q)$, we output $F = \{(p, q), (p, r), (q, r)\}$. See Figure 4(a). Observe that p, q, and r are mutually separated by a distance of at least $R \geq \Delta/4$, and both $\mathrm{diam}((\sigma_1 \setminus \sigma_2) \cap S)$ and $\mathrm{diam}((\sigma_2 \setminus \sigma_1) \cap S)$ are at most Δ. Since at least one pair in F must lie within the same slab, F necessarily contains an anchor pair of Σ.

Now we consider the remaining case where $S \subset (D_p \cup D_q)$. Let $P = S \cap D_p$ and $Q = S \cap D_q$. Let $\mathrm{conv}(P)$ and $\mathrm{conv}(Q)$ be the convex hulls of P and Q, respectively. Note that these hulls do not intersect. Let ℓ_1 and ℓ_2 be the two lines that are inner common tangents to $\mathrm{conv}(P)$ and $\mathrm{conv}(Q)$. Let $p_1 \in P$ (resp., $p_2 \in P$) and $q_1 \in Q$ (resp., $q_2 \in Q$) be the points lying on ℓ_1 (resp., ℓ_2). See Figure 4(b). We can compute p_1, p_2, q_1, and q_2 without computing the convex hulls explicitly by linear programming in $O(n)$ time [15, 24]. Let (p_3, p_4) be a pair of points in P such that $d(p_3, p_4) \geq \mathrm{diam}(P)/2$. Let (q_3, q_4) be a pair of points in Q such that $d(q_3, q_4) \geq \mathrm{diam}(Q)/2$. We can compute p_3, p_4, q_3, and q_4 in $O(n)$ time. Then we output $F = \{(p, q), (p_3, p_4), (q_3, q_4)\} \cup \bigcup_{i=1}^{4} (p, q_i) \cup \bigcup_{i=1}^{4} (q, p_i)$. Now we show that F contains at least one anchor pair of Σ.

Assume to the contrary that no pair of F is an anchor pair of Σ. Let $S_{12} = S \cap (\sigma_1 \setminus \sigma_2)$ and $S_{21} = S \cap (\sigma_2 \setminus \sigma_1)$. The assumption implies that S_{12} (resp., S_{21}) contains either p or q but not both. Without loss of generality, let $p \in S_{12}$ and $q \in S_{21}$. For every $1 \leq i \leq 4$, since both $d(p, q_i)$ and $d(q, p_i)$ are at least $R \geq \Delta/4$, the assumption also implies $p_i \in S_{12}$ and $q_i \in S_{21}$.

We first claim that $P \cap S_{21} \neq \emptyset$ and $Q \cap S_{12} \neq \emptyset$ under the assumption. Suppose that $P \cap S_{21} = \emptyset$. Then $S_{21} \subseteq Q$ and thus $\mathrm{diam}(S_{21}) \leq \mathrm{diam}(Q)$. Since $\mathrm{diam}(S_{21})/2 \leq \mathrm{diam}(Q)/2 \leq d(q_3, q_4)$ and $q_3, q_4 \in S_{21}$, (q_3, q_4) is an anchor pair of Σ, contradicting the assumption[1]. A similar contradiction occurs

[1]In the proof of [2], the authors claimed that $P \cap S_{21} \neq \emptyset$ and $Q \cap S_{12} \neq \emptyset$ in a similar way. However, under their definition of an anchor pair, it is not clear that the assumption $P \cap S_{21} = \emptyset$ (resp., $Q \cap S_{12} = \emptyset$) leads to a contradiction, because $\mathrm{diam}(Q)$ (resp., $\mathrm{diam}(P)$) can be much smaller than $\mathrm{diam}(S \cap \sigma_2)$ (resp., $\mathrm{diam}(S \cap \sigma_1)$).

if we assume $Q \cap S_{12} = \emptyset$.

Therefore there exist two points p' and q' such that $p' \in P \cap S_{21}$ and $q' \in Q \cap S_{12}$. Let s be the intersection of ℓ_1 and ℓ_2. Observe that $\triangle sq_1q_2 \subset \triangle p'q_1q_2 \subset \sigma_2$ as $p' \in P$ and $p', q_1, q_2 \in \sigma_2$. As q' should not be inside σ_2, q' should lie outside of $\triangle sq_1q_2$. Therefore $\triangle p_1p_2q'$ and $\overline{q_1q_2}$ intersect. See Figure 4(c). As $p_1, p_2, q' \in S_{12}$ and $q_1, q_2 \in S_{21}$, it is implied that σ_1 separates q_1 and q_2. In a symmetric way, we can show that σ_2 separates p_1 and p_2. This implies that the segments $\overline{p_1p_2}$ and $\overline{q_1q_2}$ intersect (see Figure 4(d)). But it contradicts that $D_p \cap D_q = \emptyset$. □

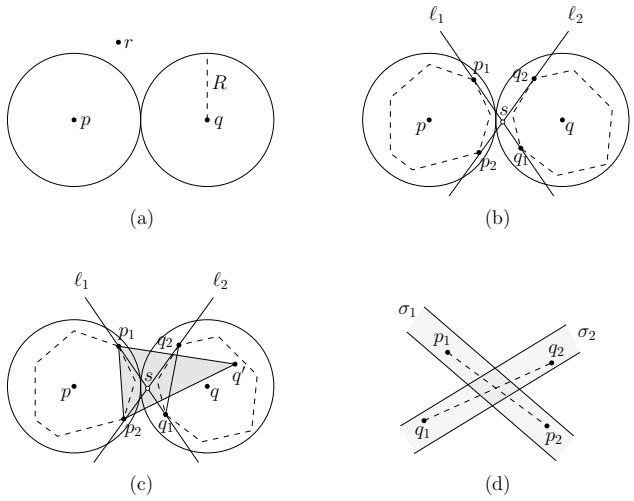


Figure 4: (a) There exists a point $r \in S \setminus (D_p \cup D_q)$. (b) The inner common tangents ℓ_1 and ℓ_2 of $conv(P)$ and $conv(Q)$, and the points p_1, p_2, q_1, q_2, and s are shown. (c) The triangle $\triangle p_1p_2q'$ and the segment $\overline{q_1q_2}$ intersect. (d) The segments $\overline{p_1p_2}$ and $\overline{q_1q_2}$ intersect.

3.2 A Linear-Time 10-Approximation Algorithm

In this section, we give a 10-approximation algorithm for the general two-line-center problem that runs in $O(n)$ time. We remark that the authors in [2] gave a 6-approximation algorithm that runs in $O(n \log n)$ time.

By Lemma 2, we can find a set of at most 11 pairs of points, at least one of which is an anchor pair of Σ^* in $O(n)$ time. For the moment, assume that we have an anchor pair (p, q) of Σ^*. Without loss of generality, assume that $p, q \in \sigma_1^*$ and $d(p, q) \geq \text{diam}((\sigma_1^* \setminus \sigma_2^*) \cap S)/4$.

Let $p_1, p_2, \ldots, p_n$ be the points of S, indexed in increasing order of their distance from ℓ_{pq}. (We do not actually sort them. The indices are used only for reference.) For three points $\alpha, \beta, \gamma \in \mathbb{R}^2$, let $\sigma(\alpha, \beta; \gamma)$ denote the slab whose center line is $\ell_{\alpha\beta}$ and one bounding line passes through γ (see Figure 5).

The following lemma implies the existence of a pair of slabs whose maximum width is at most $10w^*$, and one of which is parallel to ℓ_{pq}.

Lemma 3. *There exists a point $r \in S$ such that $\sigma(p, q; r)$ covers all points of $(\sigma_1^* \setminus \sigma_2^*) \cap S$ and $w(\sigma(p, q; r)) \leq 10w^*$. For such r,* $\text{width}(S \setminus \sigma(p, q; r)) \leq w^*$.

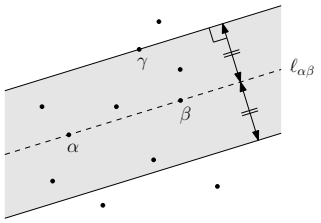


Figure 5: For three points $\alpha, \beta, \gamma \in \mathbb{R}^2$, the gray region represents $\sigma(\alpha, \beta; \gamma)$.

Proof. We prove that for every $s \in (\sigma_1^* \setminus \sigma_2^*) \cap S$, it holds that $d(\ell_{pq}, s) \leq 5w^*$. It directly implies that there exists a slab τ of width at most $10w^*$ that has ℓ_{pq} as its center line and covers $(\sigma_1^* \setminus \sigma_2^*) \cap S$. Observe that we can choose r as the first point in S touched by one of the bounding lines of τ as we move them toward ℓ_{pq} at the same speed.

Without loss of generality, we assume that p and q lie on a horizontal line. Let $D = \mathrm{diam}((\sigma_1^* \setminus \sigma_2^*) \cap S)$. Therefore, $d(p, q) \geq D/4$. Let α denote the orientation of σ_1^*. For simplicity, we assume that $0 \leq \alpha < \pi/2$. The other case can be shown in a symmetric way. Since $p, q \in \sigma_1^*$, we have $\sin \alpha \leq w^*/d(p, q) \leq 4w^*/D$. See the right triangle having $\overline{pq}$ as one of its sides in Figure 6.

Let B be the minimum bounding box of $(\sigma_1^* \setminus \sigma_2^*) \cap S$ with orientation α. Observe that every side of B has length at most D. Let o and t denote the lowest and highest vertices of B, respectively, which lie at diagonally opposite corners. Consider a horizontal slab σ whose two bounding lines pass through o and t, and let $w = w(\sigma)$. Clearly, $(\sigma_1^* \setminus \sigma_2^*) \cap S \subset \sigma$. We now show that $w \leq 5w^*$, which completes the proof.

Let b be a vertex of B other than o and t. Consider the vertical line ℓ that passes through b. Then ℓ intersects the bounding lines of σ. Observe that one intersection point lies inside σ_1^*, while the other lies outside. Let a and c denote these two intersection points, with a being the one inside σ_1^*. Then $w = \overline{ab} + \overline{bc} \leq \overline{tb} + \overline{bc} \leq w^* + \overline{bc}$. We have $\overline{bc} = \overline{ob} \cdot \sin \alpha \leq D \cdot \frac{4w^*}{D} \leq 4w^*$. Therefore, $w \leq 5w^*$. □

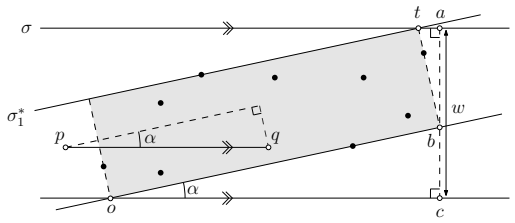


Figure 6: Black points indicate the set $(\sigma_1^* \setminus \sigma_2^*) \cap S$. The orientation of σ_1^* is denoted by α. The gray region represents B, the minimum bounding box of $(\sigma_1^* \setminus \sigma_2^*) \cap S$ with orientation α. The anchor pair (p, q) and the points o, t, a, b, c are shown. Observe that $w(\sigma) = \overline{ab} + \overline{bc}$.

Naive Approach and Idea By Lemma 3, there exists a point $r \in S$ such that both $w(\sigma(p, q; r))$ and $\mathrm{width}(S \setminus \sigma(p, q; r))$ are at most $10w^*$. We now give

a brief idea of using a binary search to find such a point r.

Consider the slab $\sigma(p, q; p_i)$ and the set $S \setminus \sigma(p, q; p_i)$ for $1 \leq i \leq n$. Observe that as i increases, the width $w(\sigma(p, q; p_i))$ increases, while $\text{width}(S \setminus \sigma(p, q; p_i))$ decreases[2]. Using this monotonicity, we can find an index i that minimizes the unimodal function $\max\{w(\sigma(p, q; p_i)), \text{width}(S \setminus \sigma(p, q; p_i))\}$.

More specifically, at each step of the binary search over a range of indices $[s, e] \subseteq [1, n]$, we compute the median index m and evaluate $w(\sigma(p, q; p_m))$ and $\text{width}(S \setminus \sigma(p, q; p_m))$. If $w(\sigma(p, q; p_m)) \geq \text{width}(S \setminus \sigma(p, q; p_m))$, we set the range to $[s, m]$. Otherwise, we set it to $[m, e]$. Let j be the index obtained by the binary search. Observe that j minimizes $\max\{w(\sigma(p, q; p_i)), \text{width}(S \setminus \sigma(p, q; p_i))\}$ over all $1 \leq i \leq n$. By Lemma 3, we have $\max\{w(\sigma(p, q; p_j)), \text{width}(S \setminus \sigma(p, q; p_j))\} \leq 10w^*$.

Note that the time required in each step is dominated by computing $\text{width}(S \setminus \sigma(p, q; p_m))$, which takes $O(n \log n)$ time [26]. Therefore, the overall running time is $O(n \log^2 n)$.

To reduce the time complexity, we avoid computing $width(S \setminus \sigma(p, q; p_m))$ from scratch at every step by sacrificing exactness. We remark that the authors in [2] presented an $O(n \log n)$-time binary search algorithm by reducing the time for each step to $O(n)$. We present an $O(n)$-time binary search algorithm in which each step takes time linear in the size of the current range of indices, even when the points are not sorted. Our algorithm uses the following incremental algorithm that computes a 6-approximation of the width of a point set.

Theorem 4 (Theorem A.3 of [12])**.** *Given an online sequence of points in the plane, we can maintain a* 6*-approximation of the width with* $O(1)$ *space and update time.*

We show the following theorem. See Appendix A for the proof and full details.

Theorem 5. *In* $O(n)$ *time, we can compute a pair of slabs that forms a* 10*-approximation for the general two-line-center problem.*

3.3 An ε-certificate

Before defining an ε-certificate, we first define the notion of an *ε-expansion*. We remark that the definitions in this section are from [3].

Definition 6 (ε-expansion)**.** *(i) For an interval* $[a, b]$*, we define its* ε-expansion *to be the interval* $[a - \varepsilon(b-a)/2, b + \varepsilon(b-a)/2]$*. See Figure 7(a).*

(ii) For a slab centered on a line ℓ *with width* w*, we define its* ε-expansion *to be the slab centered on the line* ℓ *having width* $(1 + \varepsilon)w$*. See Figure 7(b).*

Now we define an ε-certificate in both $\mathbb{R}^1$ and $\mathbb{R}^2$.

Definition 7 (ε-certificate)**.** *(i) Let* P *be a set of* n *points on the real line and* $\varepsilon > 0$*. We say that a subset* $Q \subseteq P$ *is an* ε-certificate *of* P *if for any pair of intervals whose union covers* Q*, the union of their* ε*-expansions covers* P*.*

[2]Throughout this section, we refer to a non-decreasing (resp., non-increasing) function simply as an increasing (resp., decreasing) function.

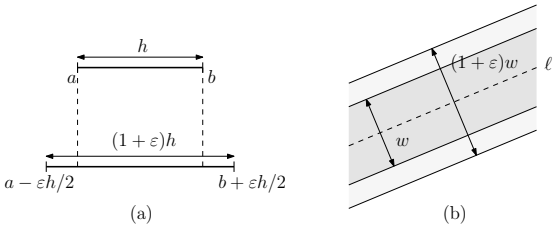


Figure 7: (a) The ε-expansion of an interval. (b) The ε-expansion of a slab.

(ii) Let P be a set of n points in the plane and $\varepsilon > 0$. We say that a subset $Q \subseteq P$ is an ε-certificate *of P if for any pair of slabs with the same width whose union covers Q, the union of their ε-expansions covers P. We emphasize that the two slabs must have the same width.*

We note that the existence of an ε-certificate of size $1/\varepsilon^{O(1)}$ was shown in [3]. By proving the following theorem, we refine the hidden constant and show how to construct such an ε-certificate in linear time.

Theorem 8. *For a set S of n points in the plane and $\varepsilon > 0$, an ε-certificate of S of size $O(1/\varepsilon^2)$ can be computed in $O(n)$ time.*

Proof. If $n \leq 1/\varepsilon^2$, then we simply take S itself as an ε-certificate of size $O(1/\varepsilon^2)$. Therefore, we only consider the case where $n > 1/\varepsilon^2$.

We first show how to compute an ε-certificate in the one-dimensional space, and then use it to handle the two-dimensional case.

One-dimensional case Let P be a set of m points on the real line. We show that an ε-certificate of P of size $O(1/\varepsilon)$ can be computed in $O(m)$ time. If $m \leq 1/\varepsilon$, then we simply take P itself as an ε-certificate of size $O(1/\varepsilon)$. Therefore, we only consider the case where $m > 1/\varepsilon$. Let I be the interval that has the leftmost and rightmost points of P as endpoints. Let D denote the length of I. We divide I into $\lceil 4/\varepsilon \rceil$ intervals of length at most $\varepsilon D/4$. Let $a_0 < a_1 < \ldots < a_{\lceil 4/\varepsilon \rceil}$ denote the endpoints of these intervals (thus, $I = [a_0, a_{\lceil 4/\varepsilon \rceil}]$). For every i, $0 \leq i < \lceil 4/\varepsilon \rceil$, we find the leftmost and rightmost points in $P \cap [a_i, a_{i+1}]$ and let R be the union of them (see Figure 8). Observe that R always contains the leftmost and rightmost points of P, and $|R| = O(1/\varepsilon)$. By using the floor function, we can determine the interval to which each input point belongs in constant time. Therefore, R can be computed in $O(m + 1/\varepsilon) = O(m)$ time.

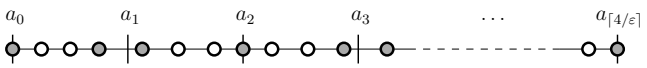


Figure 8: For a set P of points on the real line, the points in R are shown in gray.

Claim. *R is an ε-certificate of P.*

Proof. Let (I_1, I_2) be a pair of intervals whose union covers R. Without loss of generality, assume that the two intervals are interior-disjoint and that I_1 lies

to the left of I_2. We first consider the case where both I_1 and I_2 have lengths less than $D/2$. Then there exists a_i that is contained in neither I_1 nor I_2 (see Figure 9(a)). In other words, I_1 is strictly to the left of a_i, and I_2 is strictly to the right of a_i. Let $R_1 = R \cap [a_0, a_i]$ and $R_2 = R \cap [a_i, a_{\lceil 4/\varepsilon \rceil}]$. As $I_1 \cup I_2$ covers R, we have $R_1 \subset I_1$ and $R_2 \subset I_2$. Since the leftmost and rightmost points of $P \cap [a_0, a_i]$ are in R_1 and the leftmost and rightmost points of $P \cap [a_i, a_{\lceil 4/\varepsilon \rceil}]$ are in R_2, $I_1 \cup I_2$ covers P. (In this case, no expansion is needed.)

Now, consider the case where I_1 or I_2 has length at least $D/2$. Without loss of generality, assume that $|I_1| \geq D/2$. Let j be the smallest index such that $a_j \notin I_1$ (see Figure 9(b)). Note that I_1 is strictly left to a_j. Since the leftmost and rightmost points of $P \cap [a_j, a_{\lceil 4/\varepsilon \rceil}]$ are in R, I_2 contains all the points in $P \cap [a_j, a_{\lceil 4/\varepsilon \rceil}]$. Now it suffices to show that the ε-expansion of I_1 includes $[a_0, a_j)$. Observe that the right endpoint of I_1 lies in $[a_{j-1}, a_j)$. By the ε-expansion, the right endpoint shifts by $\varepsilon|I_1|/2$ to the right. By our assumption, we have $\varepsilon|I_1|/2 \geq \varepsilon D/4$. By definition, $a_i - a_{i-1} \leq \varepsilon D/4$ for every $i = 1, \ldots, \lceil 4/\varepsilon \rceil$. Therefore the ε-expansion of I_1 contains $[a_0, a_j]$. $\lhd$

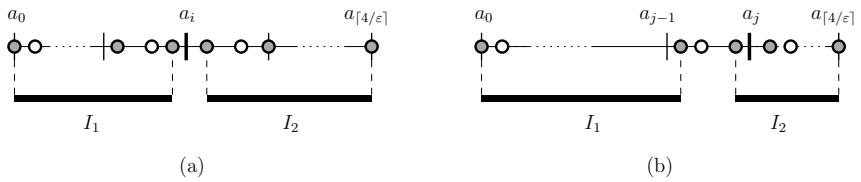


Figure 9: The points in R are shown in gray. (a) Both $|I_1|$ and $|I_2|$ are less than $D/2$. (b) $|I_1| \geq D/2$.

Two-dimensional case We now show how to compute an ε-certificate of S of size $O(1/\varepsilon^2)$ in $O(n)$ time. Recall that $w^* = \max\{w(\sigma_1^*), w(\sigma_2^*)\}$ where (σ_1^*, σ_2^*) is an optimal pair of slabs for the general two-line-center problem for S.

We first compute a pair (λ_1, λ_2) of slabs with the same width such that their union covers S and their widths are at most $10w^*$. This can be done in $O(n)$ time by Theorem 5. Let $\widetilde{w} = w(\lambda_1) = w(\lambda_2)$, that is $\widetilde{w} \leq 10w^*$. For a small constant $c > 0$ that will be specified later, let $\delta = c\varepsilon$. Let L_1 be a maximal set of equally spaced parallel lines contained in λ_1 such that $d(\ell, \ell') = \delta\widetilde{w}$ for every two consecutive lines ℓ and ℓ' in L_1. See Figure 10. We define L_2 symmetrically for λ_2. Let $L = L_1 \cup L_2$. Observe that $|L| = O(1/\delta) = O(1/\varepsilon)$.

Let S' be the point set obtained by projecting each point in S onto its closest line in L. For each $\ell \in L$, let $S'(\ell) \subseteq S'$ denote the set of projected points on ℓ. See Figure 11(a). Let $Q'(\ell) \subseteq S'(\ell)$ be an ε-certificate of $S'(\ell)$, obtained by applying the one-dimensional construction described above (treating ℓ as a one-dimensional space). See Figure 11(b). Finally, let $Q' = \bigcup_{\ell \in L} Q'(\ell)$, and let Q be the set of original points in S corresponding to Q'. See Figure 11(c). Observe that $Q \subseteq S$ and $|Q| = O(1/\varepsilon^2)$, since $|L| = O(1/\varepsilon)$ and $|Q'(\ell)| = O(1/\varepsilon)$ for each $\ell \in L$.

We can compute S' in $O(n)$ time using the floor function. It takes $O(\sum_{\ell \in L} |S'(\ell)|) = O(n)$ time to compute $Q'(\ell)$ for all $\ell \in L$. Therefore, we can compute Q in $O(n + 1/\varepsilon)$ time, which is $O(n)$ by the assumption that $n > 1/\varepsilon^2$. Now we show that Q is an ε-certificate of S.

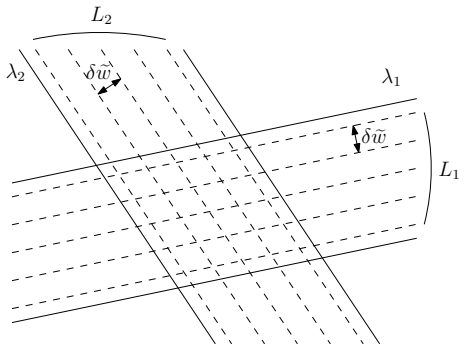


Figure 10: The sets L_1 and L_2 of lines are shown as described above.

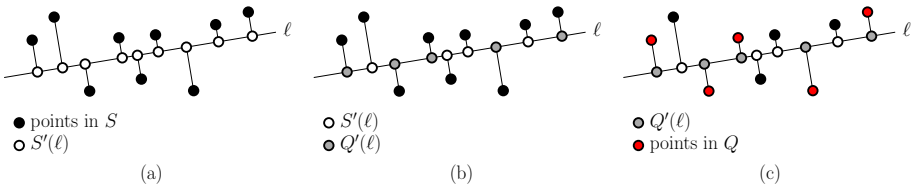


Figure 11: A rough illustration of the construction of Q from S with respect to $\ell \in L$.

Claim. *Q is an ε-certificate of S.*

Proof. Our proof follows along the same lines as the proof in [3]. Let (σ_1, σ_2) be a pair of slabs with the same width whose union covers Q. Let (σ_1', σ_2') be the pair of slabs obtained by applying a $2\delta\widetilde{w}$ additive expansion to σ_1 and σ_2, respectively. That is, $w(\sigma_1') = w(\sigma_1) + 2\delta\widetilde{w}$, and σ_1' and σ_1 have the same center line. The same holds for σ_2'. For a point q in Q, let q' denote its corresponding point in Q'. Observe that $d(q, q') \leq \delta\widetilde{w}$ for every $q \in Q$. Therefore, $Q' \subset \sigma_1' \cup \sigma_2'$ (see Figure 12(a)).

Let (τ_1', τ_2') be the pair of slabs obtained by taking the δ-expansions of σ_1' and σ_2', respectively. For each line $\ell \in L$, observe that the segment $\ell \cap \tau_1'$ is the δ-expansion of $\ell \cap \sigma_1'$. Similarly, $\ell \cap \tau_2'$ is the δ-expansion of $\ell \cap \sigma_2'$. As the union of $\ell \cap \sigma_1'$ and $\ell \cap \sigma_2'$ covers $Q'(\ell)$, the union of $\ell \cap \tau_1'$ and $\ell \cap \tau_2'$ covers $S'(\ell)$. This leads to $S' \subset \tau_1' \cup \tau_2'$ (see Figure 12(b)).

Let (τ_1, τ_2) be the pair of slabs obtained by applying a $2\delta\widetilde{w}$ additive expansion to τ_1' and τ_2', respectively. For a point p in S, let p' denote its corresponding point in S'. Observe that $d(p, p') \leq \delta\widetilde{w}$ for every $p \in S$. Therefore, $S \subset \tau_1 \cup \tau_2$ (see Figure 12(c)).

It remains to prove that $w(\tau_1) \leq (1+\varepsilon) \cdot w(\sigma_1)$ and $w(\tau_2) \leq (1+\varepsilon) \cdot w(\sigma_2)$. We only prove this for τ_1. Let $r = w(\sigma_1)$. Then, $w(\tau_1) = (r + 2\delta\widetilde{w})(1+\delta) + 2\delta\widetilde{w}$.

As $\tau_1 \cup \tau_2$ covers S, it holds that $w^* \leq (r + 2\delta\widetilde{w})(1 + \delta) + 2\delta\widetilde{w}$. Since $\widetilde{w} \leq 10w^*$, it follows that $w^* \leq (r + 20\delta w^*)(1 + \delta) + 20\delta w^*$. Therefore, we have $w^*(1 - 40\delta - 20\delta^2) \leq r(1 + \delta)$. For a sufficiently small δ, it holds that $w^*/2 \leq r(1 + \delta)$. Therefore, $w^* \leq 4r$.

Now, we rearrange $(r + 2\delta\widetilde{w})(1 + \delta) + 2\delta\widetilde{w}$ as $r(1 + \delta) + 2\delta\widetilde{w}(2 + \delta)$. Since $\widetilde{w} \leq 10w^* \leq 40r$, we have $r(1 + \delta) + 2\delta\widetilde{w}(2 + \delta) \leq r(1 + \delta) + 80\delta(2 + \delta)r =$

$(1 + 161\delta + 80\delta^2)r$. This is at most $(1 + \varepsilon)r$ for sufficiently small $c > 0$ (e.g., $c = 1/200$). ◁

This completes the proof of Theorem 8. □

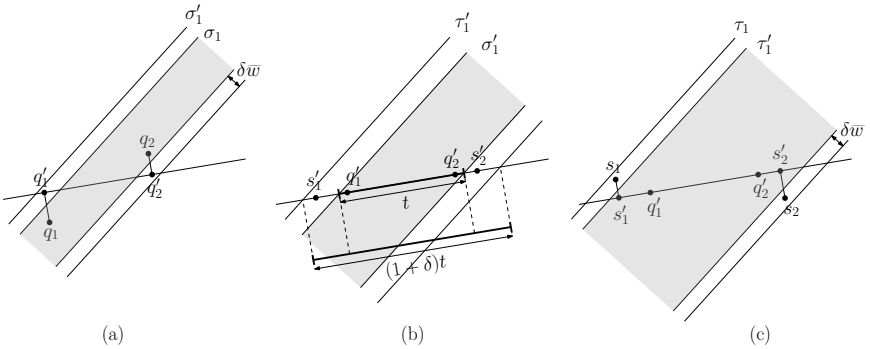


Figure 12: A rough illustration of the construction of τ_1 from σ_1.

We show that Theorem 8 directly gives a $(1 + \varepsilon)$-approximation algorithm that runs in $O(n + \text{poly}(1/\varepsilon))$ time for each of the four problems.

Theorem 9. *For each of the four problems whose optimal solution can be found in $T(n)$ time, we can compute a $(1+\varepsilon)$-approximate solution in $O(n+T(1/\varepsilon^2))$ time.*

Proof. We first compute an ε-certificate Q of S. By Theorem 8, this can be done in $O(n)$ time and $|Q| = O(1/\varepsilon^2)$.

Consider any of the four problems. Let w^* denote the maximum width of an optimal pair for S. Let $T(n)$ denote the running time of an exact algorithm for finding an optimal pair. Such a polynomial-time exact algorithm exists for each of the four problems (see Table 1). In $T(|Q|) = T(1/\varepsilon^2)$ time, we find an optimal pair (σ_1, σ_2) of slabs for Q. Without loss of generality, we assume that σ_1 and σ_2 have the same width (otherwise, we expand the smaller one). Since $Q \subseteq S$, it follows that $w(\sigma_1) = w(\sigma_2) \leq w^*$.

Let (σ_1', σ_2') be the pair of slabs obtained by taking the ε-expansions of σ_1 and σ_2, respectively. By the definition of an ε-expansion, we have $w(\sigma_1') = w(\sigma_2') \leq (1 + \varepsilon)w^*$, and the pair (σ_1', σ_2') satisfies the orientation restriction of the problem. Furthermore, by the definition of an ε-certificate, $\sigma_1' \cup \sigma_2'$ covers S. Therefore, (σ_1', σ_2') forms a $(1 + \varepsilon)$-approximate solution. □

For the general two-line-center problem, $T(n) = O(n^2 \log^2 n)$ [22]. By Theorem 9, we obtain a $(1 + \varepsilon)$-approximation algorithm running in $O(n + (1/\varepsilon^4)\log^2(1/\varepsilon))$ time. In Section 5, however, we present another algorithm with an improved running time. For the other variants as well, we present $(1+\varepsilon)$-approximation algorithms with better running times than those obtained by applying Theorem 9 with a known exact algorithm as $T(n)$ (see Table 1).

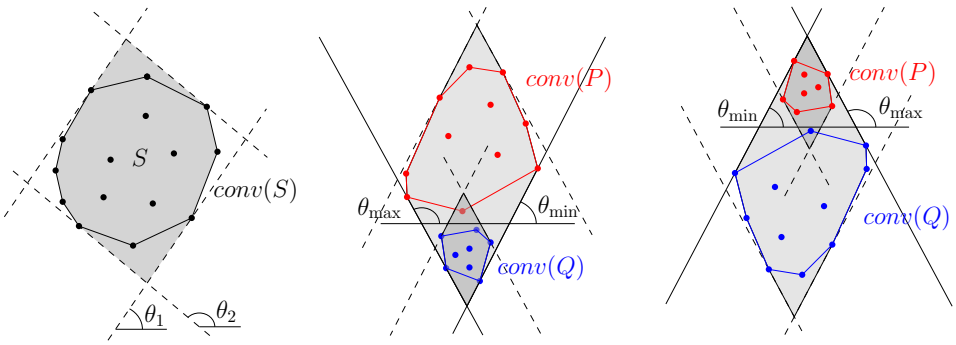
4 One Fixed Orientation

By rotating the coordinate system, we assume without loss of generality that the given orientation θ is 0. Throughout this section, we denote by (σ_1^*, σ_2^*) an optimal pair of slabs for the one-fixed-orientation problem, where σ_1^* is horizontal. Let $w^* = \max\{w(\sigma_1^*), w(\sigma_2^*)\}$.

4.1 Orientation-constrained Width and Dominance

In this section, we recall the concept of an orientation-constrained width and present geometric properties as introduced in [6]. Let $\theta_1 \leq \theta_2$ be two orientations. For a set P of points in the plane, we define the $[\theta_1, \theta_2]$-constrained width of P as $\mathrm{width}_{[\theta_1,\theta_2]}(P) := \min_{\theta\in[\theta_1,\theta_2]} \mathrm{width}_\theta(P)$. We also define $\sigma_{[\theta_1,\theta_2]}(P) := \bigcap_{\theta\in[\theta_1,\theta_2]} \sigma_\theta(P)$ (see Figure 13(a)).

Consider two non-empty sets P and Q of points in the plane that can be separated by a horizontal line, thus $\mathrm{conv}(P) \cap \mathrm{conv}(Q) = \emptyset$. Then, there are exactly two outer common tangents. Let $\theta_{\min} \leq \theta_{\max}$ be the orientations of these lines. We say that P *dominates* Q if $\sigma_{[\theta_{\min},\theta_{\max}]}(Q) \subseteq \sigma_{[\theta_{\min},\theta_{\max}]}(P)$. See Figure 13. By construction, note that either P or Q dominates the other.



(a) (b) (c)

Figure 13: (a) For a point set S and orientations θ_1 and θ_2, the gray region represents $\sigma_{[\theta_1,\theta_2]}(S)$. (b, c) For two point sets P (red) and Q (blue), the two lines represent their outer common tangents.
(b) P dominates Q. (c) Q dominates P.

Lemma 10 ([6]). *With the above notation, suppose that P dominates Q. Then*

(i) $\mathrm{width}_{[\theta_{\min},\theta_{\max}]}(P \cup Q) = \mathrm{width}_{[\theta_{\min},\theta_{\max}]}(P)$, *and*

(ii) If $\mathrm{width}(P \cup Q) < d(\mathrm{conv}(P), \mathrm{conv}(Q))$, *then* $\mathrm{width}(P \cup Q) = \mathrm{width}_{[\theta_{\min},\theta_{\max}]}(P)$.

4.2 Decision Algorithm

In this section, we present a decision algorithm for a given parameter $\omega > 0$. The algorithm determines whether there exists a pair of slabs of width ω such

that one of them is horizontal and their union covers S. If such a pair exists, we report YES. Otherwise, we report NO.

The points $s_1, s_2, \ldots, s_n$ of S are assumed to be sorted in decreasing order of their y-coordinates. For simplicity, we assume that no two input points have the same y-coordinate. Our approach follows that of the previously known algorithm from [6] but removes the dependency on the data structures they used. As a result, we obtain an $O(n)$-time decision algorithm (after sorting), improving upon the previous $O(n \log^2 n)$ bound.

Consider a horizontal slab σ_ω of width ω, and imagine sweeping σ_ω downward over the plane. For a fixed position of σ_ω, let $S_a \subseteq S$ be the set of points strictly above σ_ω, and let $S_b \subseteq S$ be the set of points strictly below σ_ω. Referring to Figure 14, observe that S_b dominates S_a from the beginning until a certain moment, after which S_a dominates S_b until the end, assuming that both S_a and S_b are non-empty. Observe that the answer to the decision problem is YES if and only if there exists a placement of σ_ω such that $\text{width}(S \setminus \sigma_\omega) \leq \omega$ (equivalently, $\text{width}(S_a \cup S_b) \leq \omega$).

Assumption For the case where the answer is YES, we assume that there exists a placement of σ_ω such that S_a dominates S_b. The other case can be handled symmetrically by sweeping σ_ω upward from the bottom.

Notation If both S_a and S_b are non-empty, there exist two outer common tangents of them. Observe that these tangent lines do not intersect each other inside σ_ω. Let ℓ_1 denote the right tangent and ℓ_2 denote the left tangent. We denote their orientations by θ_1 and θ_2, respectively. See Figure 14(a). Let $a_1 \in S_a$ and $b_1 \in S_b$ be the points of tangency defining ℓ_1, and similarly, let $a_2 \in S_a$ and $b_2 \in S_b$ be the points of tangency defining ℓ_2.[3] If either S_a or S_b is empty, then ℓ_1 and ℓ_2 (and consequently $\{a_i, b_i, \theta_i \mid i = 1, 2\}$) are undefined.

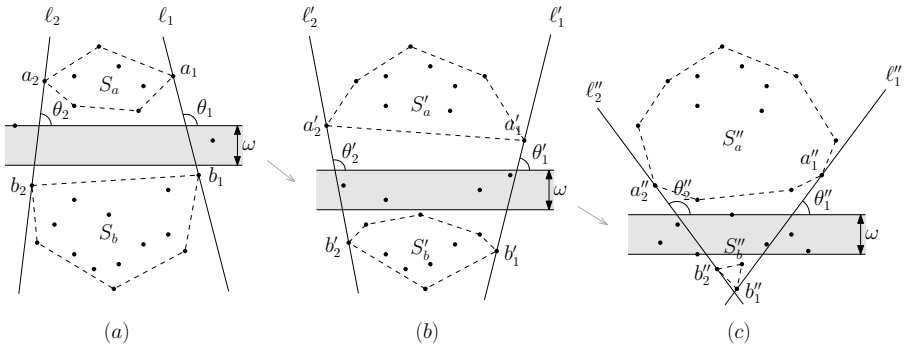


Figure 14: Three snapshots of the sweeping process for a point set. (a) S_a, S_b, and $\{\ell_i, a_i, b_i, \theta_i \mid i = 1, 2\}$ are shown according to the notation defined above. Observe that S_b dominates S_a. (b, c) We use the notations S'_a, S'_b and S''_a, S''_b (and the associated primed and double-primed variables) analogously for the two horizontal slabs positioned below. Observe that S'_a (resp., S''_a) dominates S'_b (resp., S''_b). (a–c) Observe that $\theta_1 \geq \theta'_1 \geq \theta''_1$ and $\theta_2 \leq \theta'_2 \leq \theta''_2$.

[3]There may be cases where more than one point of S_a (resp., S_b) lies on the tangent line. Our algorithm works even if one of them is chosen arbitrarily. Thus, for simplicity, we assume throughout this section that there is exactly one point of S_a (resp., S_b) on the tangent line.

Observation 11. *With the above notation, S_a dominates S_b if and only if $\theta_1 \leq \theta_2$.*

Proof. First, consider the case where $\theta_1 \leq \theta_2$. For any orientation $\theta \in [\theta_1, \theta_2]$, it holds that $\sigma_\theta(S_b) \subseteq \sigma_\theta(S_a)$ (see Figure 14(b), with a slight abuse of notation). Therefore, we have $\sigma_{[\theta_1,\theta_2]}(S_b) \subseteq \sigma_{[\theta_1,\theta_2]}(S_a)$.

Now consider the other case where $\theta_1 > \theta_2$. For any orientation $\theta \in [\theta_2, \theta_1]$, it holds that $\sigma_\theta(S_a) \subsetneq \sigma_\theta(S_b)$ (see Figure 14(a)). Therefore, we have $\sigma_{[\theta_2,\theta_1]}(S_a) \subsetneq \sigma_{[\theta_2,\theta_1]}(S_b)$. □

4.2.1 Overview of the Algorithm

In this section, we give an overview of the data structures maintained throughout the algorithm and explain how the main loop proceeds. We begin by recalling a data structure for maintaining the convex hull of a point set under a special sequence of insertions and deletions.

Lemma 12 ([29])**.** *We can dynamically maintain the convex hull of a set P of points in the plane to support each* window-sliding update *(i.e., either insert a point below all points of P or delete the topmost point of P) in $O(1)$ amortized time. The convex hull can be explicitly reported in $O(h)$ time, where h is the number of vertices of the convex hull.*

We note that in the above data structure, the neighbors of any given vertex on the convex hull can be found in $O(1)$ time. Although it is not explicitly mentioned in [29], it can be achieved since each point p (implicitly) maintains two pointers to its left and right neighbors, respectively, if p is a vertex of the current convex hull (see Sections 3.1 and 4 of [29]).

Observe that as σ_ω moves downward, a point is inserted into S_a below all existing points of S_a, or the topmost point of S_b is deleted from S_b. We refer to each such moment as *an event*. Each event corresponds to a window-sliding update of either S_a or S_b, and we have at most $2n$ events.

Data structures and Main loop As σ_ω moves downward, we maintain the following using the notation defined above:

- the dynamic data structures for $\mathrm{conv}(S_a)$ and $\mathrm{conv}(S_b)$ (from Lemma 12),
- the tangent lines ℓ_1 and ℓ_2, with their corresponding points a_1, b_1, a_2, and b_2,
- the two antipodal pairs of S_a with respect to the orientations θ_1 and θ_2, respectively.[4]

Initially, we start with a position of σ_ω such that S_a contains only s_1, the topmost point of S. If the corresponding set S_b is empty, we stop the algorithm and report YES. Otherwise, we initialize the data structures for $\mathrm{conv}(S_a)$ and $\mathrm{conv}(S_b)$, respectively. Then we find ℓ_1 and ℓ_2, together with $\{a_i, b_i, \theta_i \mid i = 1, 2\}$. Note that the two antipodal pairs of S_a with respect to θ_1 and θ_2 are both (s_1, s_1).

[4]There may be more than one antipodal pair of S_a with respect to θ_1 (resp., θ_2). Our algorithm works even if one of them is chosen arbitrarily. Thus, for simplicity, we assume throughout this section that there is exactly one antipodal pair with respect to θ_1 (resp., θ_2).

In the main loop of our decision algorithm, we do the following as σ_ω moves downward. If S_b dominates S_a, we just update the above data structures. Once S_a begins to dominate S_b, we include, for every next event, an additional step that tests whether $\text{width}_{[\theta_1,\theta_2]}(S_a) \leq \omega$, using the maintained data structures. If this condition holds, we stop the algorithm and report YES; otherwise, we proceed the algorithm.

If S_b becomes empty, we set $\theta_1 = 0$ and $\theta_2 = \pi$. We then test whether $\text{width}_{[\theta_1,\theta_2]}(S_a) = \text{width}(S_a) \leq \omega$. If this condition holds, we report YES; otherwise, we report NO.

The correctness of this algorithm, assuming that all data structures are maintained properly and the test $\text{width}_{[\theta_1,\theta_2]}(S_a) \leq \omega$ is evaluated correctly, is guaranteed by the following lemma.

Lemma 13. *For any placement of σ_ω where S_a dominates S_b, we have* $\text{width}(S_a \cup S_b) \leq \omega$ *if and only if* $\text{width}_{[\theta_1,\theta_2]}(S_a) \leq \omega$.

Proof. First, observe that $\omega < d(\text{conv}(S_a), \text{conv}(S_b))$ as S_a and S_b are separated by a horizontal slab of width ω. If $\text{width}(S_a \cup S_b) \leq \omega$, then $\text{width}(S_a \cup S_b) = \mathit{width}_{[\theta_1,\theta_2]}(S_a)$ by the above observation and Lemma 10(ii). Therefore, we have $\mathit{width}_{[\theta_1,\theta_2]}(S_a) \leq \omega$.

Conversely, if $\mathit{width}_{[\theta_1,\theta_2]}(S_a) \leq \omega$, then $\mathit{width}_{[\theta_1,\theta_2]}(S_a \cup S_b) \leq \omega$ by Lemma 10(i). By definition, $\text{width}(S_a \cup S_b) \leq \mathit{width}_{[\theta_1,\theta_2]}(S_a \cup S_b)$. Therefore, we have $\text{width}(S_a \cup S_b) \leq \omega$. □

4.2.2 Maintaining Data Structures and Testing the Width Condition

In this section, we describe how to maintain the data structures and test the condition $\text{width}_{[\theta_1,\theta_2]}(S_a) \leq \omega$ for each event that occurs as σ_ω moves downward.

We begin by proving Lemma 14, which motivates the part of the algorithm that tests whether $\text{width}_{[\theta_1,\theta_2]}(S_a) \leq \omega$. Let σ_ω be a horizontal slab of width ω where S_a dominates S_b. Recall that θ_1 and θ_2 are the orientations of the right and left outer tangent lines, respectively. By Observation 11, we have $\theta_1 \leq \theta_2$. Now, let σ'_ω be another horizontal slab of width ω, positioned below σ_ω (the two slabs may partially overlap). We use the notations S'_a, θ'_1, and θ'_2 analogously for σ'_ω. Then, it always holds that $[\theta_1, \theta_2] \subseteq [\theta'_1, \theta'_2]$.

Lemma 14. *With the above notation, assume that $\mathit{width}_{[\theta_1,\theta_2]}(S_a) > \omega$. Then, $\mathit{width}_{[\theta'_1,\theta'_2]}(S'_a) \leq \omega$ if and only if $\mathit{width}_{[\theta'_1,\theta_1]}(S'_a) \leq \omega$ or $\mathit{width}_{[\theta_2,\theta'_2]}(S'_a) \leq \omega$.*

Proof. Since $[\theta_1, \theta_2] \subseteq [\theta'_1, \theta'_2]$, we can partition the range $[\theta'_1, \theta'_2]$ into three subranges: $[\theta'_1, \theta_1]$, $[\theta_1, \theta_2]$, and $[\theta_2, \theta'_2]$. Let $a = \mathit{width}_{[\theta'_1,\theta_1]}(S'_a)$, $b = \mathit{width}_{[\theta_1,\theta_2]}(S'_a)$, and $c = \mathit{width}_{[\theta_2,\theta'_2]}(S'_a)$. By definition, $\mathit{width}_{[\theta'_1,\theta'_2]}(S'_a) = \min\{a, b, c\}$.

By the assumption and since $S'_a \supseteq S_a$, we have $b \geq \mathit{width}_{[\theta_1,\theta_2]}(S_a) > \omega$. If $\mathit{width}_{[\theta'_1,\theta'_2]}(S'_a) \leq \omega$, then a or c should be at most ω. Conversely, if $a \leq \omega$ or $c \leq \omega$, it follows that $\mathit{width}_{[\theta'_1,\theta'_2]}(S'_a) \leq \omega$. □

We first note that the data structures (from Lemma 12) for $\text{conv}(S_a)$ and $\text{conv}(S_b)$ can be maintained in $O(n)$ total time, since each event corresponds to a window-sliding update.

We now describe how to handle each event depending on its type: either an insertion into S_a or a deletion from S_b.

We first observe that if the point being inserted or deleted lies strictly between ℓ_1 and ℓ_2, then nothing changes except the convex hulls (see Figure 15). Therefore, we assume that the point corresponding to an event does not lie strictly between ℓ_1 and ℓ_2.

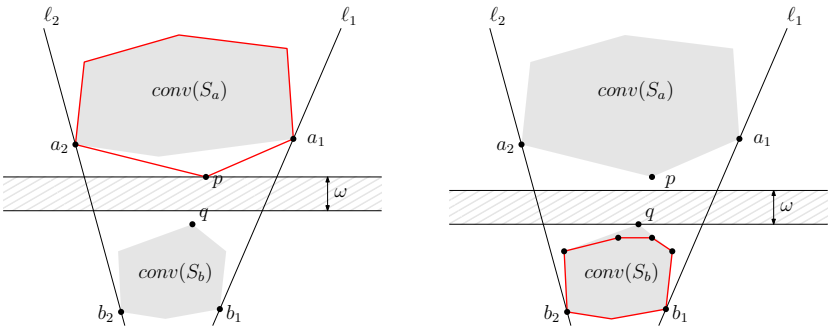


Figure 15: (a) p is inserted into S_a. (b) q is deleted from S_b. The red polygons show the convex hulls immediately after the events. Observe that the tangent lines ℓ_1 and ℓ_2 remain unchanged.

Insertion into S_a Let p be the point inserted into S_a, and assume that p lies to the right of ℓ_1 (see Figure 16). The case where p lies to the left of ℓ_2 can be handled symmetrically.

Tangent searching By our assumption, ℓ_2 remains unchanged, while ℓ_1 changes. Let ℓ_1' denote the new tangent line, and let θ_1' denote its orientation. Let $S_a' = S_a \cup \{p\}$. Analogously, let $a_1' \in S_a'$ and $b_1' \in S_b$ be the points lying on ℓ_1'. Note that $a_1' = p$. We find b_1' by traversing $\text{conv}(S_b)$ clockwise from b_1; we refer to this as the *insertion-type tangent searching procedure*. This procedure takes $O(1 + k)$ time, where k is the number of vertices on $\text{conv}(S_b)$ strictly between b_1 and b_1'. We say that these vertices are *involved* in the procedure.

Antipodal pair searching Now we find the antipodal pairs of S_a' with respect to θ_1' and θ_2 (since ℓ_2 does not change). Let q be the point such that (a_1, q) is the antipodal pair of S_a with respect to θ_1.

For the orientation θ_1', observe that $a_1'(= p)$ is one point of the antipodal pair of S_a'. We can find the other point by traversing $\text{conv}(S_a')$ clockwise from q. Let q' denote the point such that (a_1', q') is the antipodal pair of S_a' with respect to θ_1'. We refer to this as the *insertion-type antipodal-pair searching procedure*, which takes $O(1 + k')$ time, where k' is the number of vertices on $\text{conv}(S_a')$ (equivalently, on $\text{conv}(S_a)$) strictly between q and q'. We say that these vertices are *involved* in the procedure.

For the orientation θ_2, observe that a_2 is always one point of the antipodal pairs for both S_a and S_a'. Let r be the point such that (a_2, r) forms an antipodal pair of S_a with respect to θ_2 (see Figure 16(b)). Let r' be the point such that (a_2, r') forms an antipodal pair of S_a' with respect to θ_2. Then $r' = r$ if p lies to the left of the line of orientation θ_2 passing through r, and $r' = p$ otherwise.

Testing the width condition If S_a' does not dominate S_b, there is nothing left to do. This can be easily detected by Observation 11. Otherwise, we need

to test whether $\mathrm{width}_{[\theta_1',\theta_2]}(S_a') \leq \omega$. By Lemma 14, it suffices to determine whether $\mathrm{width}_{[\theta_1',\theta_1]}(S_a') \leq \omega$. We do this by following the antipodal pairs of S_a' from the one corresponding to θ_1' to that of θ_1. Note that p is always one point of the antipodal pairs of S_a' for all orientations between θ_1' and θ_1, since p lies to the right of ℓ_1. Hence, (p, q) is the antipodal pair of S_a' with respect to θ_1, and (p, q') is the one with respect to θ_1' as explained above. By traversing $\mathrm{conv}(S_a')$ counterclockwise starting from q', we can visit all antipodal pairs corresponding to the range of orientations $[\theta_1', \theta_1]$ in order. Following them sequentially allows us to find an orientation $\theta \in [\theta_1', \theta_1]$ that minimizes $\mathrm{width}_\theta(S_a')$. This procedure takes $O(1 + k')$ time, where, as before, k' is the number of vertices on $\mathrm{conv}(S_a')$ (equivalently, on $\mathrm{conv}(S_a)$) that lie strictly between q and q'. Note that this is bounded by the time required for the insertion-type antipodal-pair searching procedure.

The total time for handling point insertions as σ_ω moves downward depends on the two procedures: the insertion-type tangent searching procedure and the insertion-type antipodal-pair searching procedure. We conclude that it takes $O(n)$ time in total by showing that each input point can be involved at most once in each of the procedures.

Lemma 15. *Each point of S can be involved in the insertion-type tangent searching procedure and the insertion-type antipodal-pair searching procedure at most once, respectively, in the entire algorithm.*

Proof. We first show that each input point is involved in the insertion-type tangent searching procedure at most once. For a point p inserted into S_a, recall that $b_1 \in S_b$ is the original point of tangency on ℓ_1, and $b_1' \in S_b$ is the new point of tangency on ℓ_1' (see Figure 16(a)). We use the following observation: when traversing the points on $\mathrm{conv}(S_b)$ in clockwise order from b_1 to b_1', their y-coordinates decrease.

Let s be a point on $\mathrm{conv}(S_b)$ that lies strictly between b_1 and b_1', that is, $y(b_1) > y(s) > y(b_1')$. Assume to the contrary that s is involved again in the tangent searching procedure when another point t is inserted after p. Let u_1 and u_1' denote the original and new points of tangency defined analogously for this insertion. By the above observation, we have $y(u_1) > y(s) > y(u_1')$. Moreover, since the new point of tangency is found by traversing the convex hull in clockwise order, it holds that $y(b_1) > y(b_1') \geq y(u_1) > y(u_1')$. This contradicts that $y(b_1) > y(s) > y(b_1')$.

In a similar way, we can show that each input point is involved in the insertion-type antipodal-pair searching procedure at most once. Recall that q and q' are the points such that (a_1, q) is the antipodal pair of S_a with respect to θ_1 and (a_1', q') is the one of S_a' with respect to θ_1' (see Figure 16(a)). We use a similar observation: when traversing the points on $\mathrm{conv}(S_a)$ in clockwise order from q to q', their y-coordinates increase.

Let s be a point on $\mathrm{conv}(S_a)$ that lies strictly between q and q', that is, $y(q) < y(s) < y(q')$. Assume to the contrary that s is involved again in the antipodal-pair searching procedure when another point t is inserted after p. Analogously, let v and v' denote the points such that v is a point of the original antipodal pair and v' is a point of the new antipodal pair for this insertion. By the above observation, we have $y(v) < y(s) < y(v')$. Moreover, since a point of the new antipodal pair is found by traversing the convex hull in clockwise

order, it holds that $y(q) < y(q') \leq y(v) < y(v')$. This contradicts $y(q) < y(s) < y(q')$. □

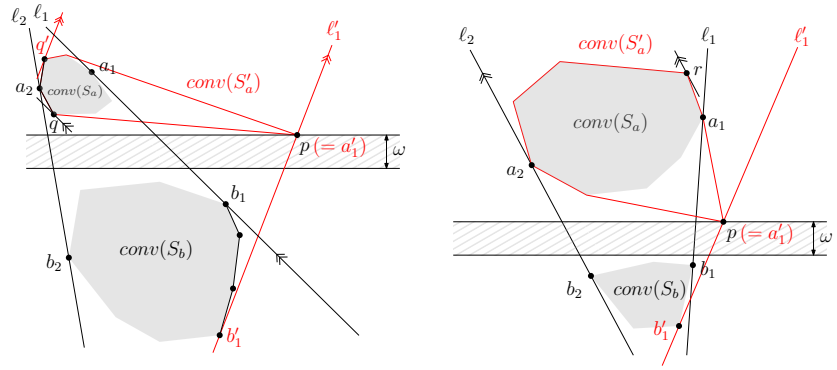


Figure 16: Two cases of insertions. In each figure, the changes after the insertion are shown in red: $\mathrm{conv}(S'_a)$, a'_1, b'_1, and ℓ'_1. In (a), observe that p becomes a point of the antipodal pair of S'_a with respect to θ_2. In contrast, in (b), the antipodal pair of S'_a with respect to θ_2 remains (a_2, r) even after inserting p. In (a), two points are involved in the tangent searching procedure, and one point is involved in the antipodal-pair searching procedure.

In a symmetric way, we handle the deletion of a point from S_b.

Deletion from S_b Let p be the point deleted from S_b. As we only consider the case where p does not lie strictly between ℓ_1 and ℓ_2, p should lie on ℓ_1 or ℓ_2. We assume that p lies on ℓ_1, that is $p = b_1$ (see Figure 17). The case where p lies on ℓ_2 can be handled symmetrically.

Tangent searching By our assumption, ℓ_2 remains unchanged, while ℓ_1 changes. Let ℓ'_1 denote the new tangent line, and let θ'_1 denote its orientation. Define $S'_b = S_b \setminus \{p\}$. Analogously, let $a'_1 \in S_a$ and $b'_1 \in S'_b$ be the points lying on ℓ'_1.

We find a'_1 and b'_1 as follows. Let t be the counterclockwise neighbor of p on $\mathrm{conv}(S_b)$. Starting from t, we traverse $\mathrm{conv}(S'_b)$ clockwise until we find a point v_1 on $\mathrm{conv}(S'_b)$ such that $\overline{a_1 v_1}$ is tangent to $\mathrm{conv}(S'_b)$ at v_1. If $\overline{a_1 v_1}$ is also tangent to $\mathrm{conv}(S_a)$ at a_1, we are done. Otherwise, we set $u = a_1$ and $v = v_1$. We then move u to its clockwise neighbor on $\mathrm{conv}(S_a)$, and update v so that $\overline{uv}$ becomes tangent to $\mathrm{conv}(S'_b)$ at v. The point v can be updated by traversing $\mathrm{conv}(S'_b)$ clockwise from its previous position. We repeat this process until $\overline{uv}$ becomes tangent to both $\mathrm{conv}(S_a)$ at u and $\mathrm{conv}(S'_b)$ at v. The final positions of u and v are a'_1 and b'_1.

We call the above the *deletion-type tangent searching procedure*, which takes $O(1 + k_1 + k_2)$ time, where k_1 is the number of points on $\mathrm{conv}(S_a)$ strictly between a_1 and a'_1, and k_2 is the number of points on $\mathrm{conv}(S'_b)$ strictly between t and b'_1. We say that these $k_1 + k_2$ points are *involved* in the procedure.

Antipodal pair searching Now we find the antipodal pairs of S_a with respect to θ'_1. Observe that the antipodal pairs of S_a with respect to θ_2 remain

unchanged as ℓ_2 does not change. Let q be the point such that (a_1, q) is the antipodal pair of S_a with respect to θ_1 (see Figure 17).

For orientation θ_1', observe that a_1' is one point of the antipodal pair of S_a. We find the other point by traversing $\mathrm{conv}(S_a)$ clockwise from q. Let q' denote the new point such that (a_1', q') is the antipodal pair of S_a with respect to θ_1'. We refer to this as the *deletion-type antipodal-pair searching procedure*, which takes $O(1 + k')$ time, where k' is the number of vertices on $\mathrm{conv}(S_a)$ strictly between q and q'. We say that these vertices are *involved* in the procedure.

Testing the width condition If S_a does not dominate S_b', there is nothing left to do. This can be easily detected by Observation 11. Otherwise, we need to test whether $\mathrm{width}_{[\theta_1', \theta_2]}(S_a) \leq \omega$. By Lemma 14, it suffices to determine whether $\mathrm{width}_{[\theta_1', \theta_1]}(S_a) \leq \omega$. We do this by following the antipodal pairs of S_a from the one corresponding to θ_1' to that of θ_1. Recall that (a_1', q') and (a_1, q) are the antipodal pairs of S_a with respect to θ_1' and θ_1, respectively. By traversing $\mathrm{conv}(S_a)$ clockwise starting from a_1 and q simultaneously, we can visit all antipodal pairs corresponding to the range of orientations $[\theta_1', \theta_1]$. Following them sequentially allows us to find an orientation $\theta \in [\theta_1', \theta_1]$ that minimizes $\mathrm{width}_\theta(S_a)$. This takes $O(1 + k_1 + k')$ time, where, as before, k_1 is the number of points on $\mathrm{conv}(S_a)$ strictly between a_1 and a_1', and k' is the number of vertices on $\mathrm{conv}(S_a)$ strictly between q and q'. Therefore, this is bounded by the total time required by the deletion-type tangent searching procedure and the deletion-type antipodal-pair searching procedure.

The total time for handling point deletions as σ_ω moves downward depends on the two procedures: the deletion-type tangent searching procedure and the deletion-type antipodal-pair searching procedure. We conclude that it takes $O(n)$ time in total by showing that each input point can be involved at most twice in each of the procedures.

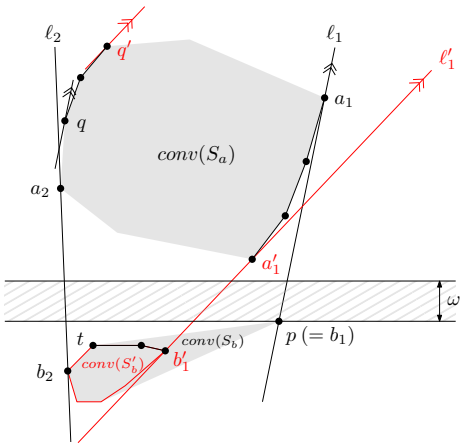


Figure 17: Changes after the deletion of p are shown in red: $\mathrm{conv}(S_b')$, a_1', b_1', and ℓ_1'. Observe that (a_1', q') becomes the antipodal pair of S_a with respect to θ_1' after the deletion. Observe that three points $(k_1 = 2, k_2 = 1)$ are involved in the deletion-type tangent searching procedure and one point is involved in the deletion-type antipodal-pair searching procedure.

Lemma 16. *Each point of S can be involved in the deletion-type tangent searching procedure and the deletion-type antipodal-pair searching procedure at most twice, respectively, in the entire algorithm.*

Proof. For a point p deleted from S_b, recall that k_1 denotes the number of points on $\mathrm{conv}(S_a)$ strictly between a_1 and a_1', and k_2 is the number of points on $\mathrm{conv}(S_b')$ strictly between t and b_1', where t is the counterclockwise neighbor of p on $\mathrm{conv}(S_b)$.

We first show that each input point is involved in the procedure as one of the k_1 points at most once. We use the following observation: when traversing the points on $\mathrm{conv}(S_a)$ in clockwise order from a_1 to a_1', their y-coordinates decrease. Let s be a point on $\mathrm{conv}(S_a)$ that lies strictly between a_1 and a_1', that is, $y(a_1) > y(s) > y(a_1')$. Assume to the contrary that s is involved again in the same procedure when another point t is inserted after p. Let u_1 and u_1' denote the original and new points of tangency of S_a (of that moment) defined analogously for this insertion. By the above observation, we have $y(u_1) > y(s) > y(u_1')$. Moreover, since the new point of tangency is found by traversing the convex hull in clockwise order, it holds that $y(a_1) > y(a_1') \geq y(u_1) > y(u_1')$. This contradicts that $y(a_1) > y(s) > y(a_1')$.

We now show that each input point is involved in the procedure as one of the k_2 points at most once. This follows directly from the following observation: A point involved in this procedure (as one of the k_2 points) becomes a vertex of the convex hull at the moment when p is deleted and remains on the convex hull until it is deleted during the sweeping process. Once a point is involved in this procedure, it cannot be involved again, since after being deleted from the convex hull, it cannot appear on it again.

It now remains to show that each input point is involved in the deletion-type antipodal-pair searching procedure (i.e., as one of the k' points) at most once. Recall that (a_1, q) and (a_1', q') are the antipodal pairs of S_a with respect to θ_1 and θ_1', respectively. We use a similar observation with the case for k_1: when traversing the points on $\mathrm{conv}(S_a)$ in clockwise order from q to q', their y-coordinates increase. Let s be a point on $\mathrm{conv}(S_a)$ that lies strictly between q and q', that is, $y(q) < y(s) < y(q')$. Assume to the contrary that s is involved again in the antipodal-pair searching procedure when another point t is inserted after p. Analogously, let v and v' denote the points such that v is a point of the original antipodal pair and v' is a point of the new antipodal pair for this insertion. By the above observation, we have $y(v) < y(s) < y(v')$. Moreover, since a point of the new antipodal pair is found by traversing the convex hull in clockwise order, it holds that $y(q) < y(q') \leq y(v) < y(v')$. This contradicts $y(q) < y(s) < y(q')$. □

Observe that the initialization of all data structures can be done in $O(n)$ time by Lemma 12. By Lemma 15 and 16, the total time to process all events and to evaluate the condition $\mathrm{width}_{[\theta_1', \theta_1]}(S_a) \leq \omega$ as σ_ω moves downward is $O(n)$. Now we conclude the following theorem.

Theorem 17. *Given a set S of n points in the plane, sorted by their y-coordinates, and a real value $\omega > 0$, we can decide in $O(n)$ time whether there exists a pair of slabs of width ω whose union covers S and one of which is horizontal.*

4.3 Optimization Algorithm

We give a brief summary of our approach to obtain Theorem 18. A complete proof is given in Appendix B. The optimality of our algorithm is given in Appendix C. We first sort the points of S so that $s_1, \ldots, s_n$ are ordered in decreasing order of their y-coordinates. Let W denote the set of all pairwise differences between the y-coordinates of points in S. Since W can be represented as a sorted matrix [16], we can identify two consecutive values $w_0, w_1 \in W$ such that $w_0 < w^* \leq w_1$ in $O(n \log n)$ time, by combining our decision algorithm described in Section 4.2 with an efficient selection algorithm for sorted matrices [17].

Observe that the sequence of changes in the sets S_a and S_b that occur as σ_w moves downward is the same for any $w_0 < w < w_1$. Using this observation, we show that w^* can be computed by tracking $width(S_a \cup S_b)$ as σ_w moves downward, for any $w_0 < w < w_1$. We track $width(S_a \cup S_b)$ by following essentially the same procedure as in the decision algorithm. The authors of [6] also use this idea, but their algorithm requires an additional data structure, which results in an extra logarithmic factor.

Theorem 18. *Given a set S of n points in the plane and an orientation θ, we can find a pair of slabs such that one of them has orientation θ and their union covers S while minimizing the maximum of their widths in $O(n \log n)$ time.*

4.4 Approximation Algorithm

Recall that in Section 3.3, we showed that an ε-certificate $Q \subseteq S$ of size $O(1/\varepsilon^2)$ can be computed in $O(n)$ time. After obtaining Q, we find a pair of slabs (σ_1, σ_2) such that their union covers Q, σ_1 has orientation θ, and the maximum of their widths is minimized. Without loss of generality, we assume that σ_1 and σ_2 have the same width (otherwise, we expand the smaller one). Since $Q \subseteq S$, it follows that $w(\sigma_1) = w(\sigma_2) \leq w^*$. Such a pair can be found in $O(|Q| \log |Q|)$ time by Theorem 18.

Let (σ_1', σ_2') be the pair of slabs obtained by taking the ε-expansions of σ_1 and σ_2, respectively. By the definition of an ε-expansion, σ_1' has orientation θ and $w(\sigma_1') = w(\sigma_2') \leq (1+\varepsilon)w^*$. Furthermore, by the definition of an ε-certificate, $\sigma_1' \cup \sigma_2'$ covers S. Therefore, (σ_1', σ_2') forms a $(1+\varepsilon)$-approximate solution.

It takes $O(n + |Q| \log |Q|)$ time to obtain (σ_1', σ_2'). Let $\mu = \min\{n, 1/\varepsilon^2\}$, then we have $|Q| = O(\mu)$. Therefore, the total running time is $O(n + \mu \log \mu)$, which can be expressed as both $O(n \log(1/\varepsilon))$ and $O(n + (1/\varepsilon^2)\log(1/\varepsilon))$. We now conclude the following theorem.

Theorem 19. *Given a set S of n points in the plane and an orientation θ, we can find a pair of slabs such that their union covers S, one of them has orientation θ, and the maximum of their widths is at most $(1+\varepsilon)w^*$ in $O(n + \mu \log \mu)$ time, where $\mu = \min\{n, 1/\varepsilon^2\}$.*

5 General Two-Line-Center

Throughout this section, we denote by $\Sigma^* = (\sigma_1^*, \sigma_2^*)$ an optimal pair of slabs for the general two-line-center problem. Let $w^* = \max\{w(\sigma_1^*), w(\sigma_2^*)\}$.

We begin by proving Lemma 20, which will be used to compute a candidate set of orientations for one of the slabs in a $(1+\varepsilon)$-approximate solution.

Lemma 20. *Let p and q be two points in $\mathbb{R}^2$, let $\widetilde{w} > 0$ be a real number, let $c \geq 1$ be a constant, and let $\varepsilon > 0$ be a real number. In $O(1/\varepsilon)$ time, we can compute a set Γ of $O(1/\varepsilon)$ orientations, such that the following holds:*
For every finite set P of points in $\mathbb{R}^2$ and for every slab σ, if
(1) *$P \subset \sigma$,* **(2)** *$p, q \in \sigma$,* **(3)** *$d(p,q) \geq \mathrm{diam}(P)/4$, and* **(4)** *$w(\sigma) \leq \widetilde{w} \leq c \cdot w(\sigma)$,*
then there exists an orientation $\gamma \in \Gamma$ and a slab σ', such that
(1) *$P \subset \sigma'$,* **(2)** *$w(\sigma') \leq (1+\varepsilon) \cdot w(\sigma)$, and* **(3)** *$\sigma'$ has the orientation γ.*

Proof. We remark that this proof is based on the approach in [2]. Without loss of generality, assume that p and q lie on a horizontal line. Let $D = d(p,q)$. Let θ be the orientation with $0 < \theta \leq \pi/2$ such that $\sin\theta = \min\{1, \widetilde{w}/D\}$. Let $\delta = c_\delta \varepsilon$ for some constant $0 < c_\delta < 1$. Let $\Gamma = \{\gamma_i = (i - \lceil \frac{1}{\delta} \rceil) \cdot \delta\theta \mid 0 \leq i \leq 2\lceil \frac{1}{\delta} \rceil\}$. Observe that Γ contains uniformly placed orientations in the range $[-\theta, \theta]$, and $|\Gamma| = O(1/\varepsilon)$. (Here we slightly abuse the notation by redefining orientations to lie in $[-\pi/2, \pi/2]$ instead of $[0, \pi]$.) In the following, we prove that Γ satisfies the conditions in the statement.

Let P be a set of points in $\mathbb{R}^2$ and let σ be a slab, where P and σ satisfy the four conditions in the statement. Let α be the orientation of σ, and let $w = w(\sigma)$. For simplicity, we assume that $0 \leq \alpha < \pi/2$. The other case can be shown in a symmetric way. Since $p, q \in \sigma$, we have $\sin\alpha \leq w/D \leq \widetilde{w}/D$ (see the right triangle having $\overline{pq}$ as one of its sides in Figure 18). Observe that if $\widetilde{w}/D < 1$, then $\sin\theta = \widetilde{w}/D$. If $\widetilde{w}/D \geq 1$, then $\sin\theta = 1$. In both cases, it holds that $\sin\alpha \leq \sin\theta$. Therefore, we have $\alpha \leq \theta$.

Let γ be the largest orientation in Γ such that $\gamma \leq \alpha$. Since $\alpha \leq \theta$, we have $\alpha - \gamma < \delta\theta$. Let R be the minimum bounding box of P with orientation α. See Figure 18. Denote by o and t the lowest and highest vertices of R, respectively. Now consider a slab σ' of orientation γ whose two bounding lines pass through o and t. Clearly, $P \subset \sigma'$. We next show that $w(\sigma') \leq (1+\varepsilon)w$, which completes the proof.

Let b be a vertex of R other than o and t. Consider the line ℓ orthogonal to σ' that passes through b. The line ℓ intersects the two bounding lines of σ'. Observe that one intersection point lies inside σ, while the other lies outside. Let a and c denote these two intersection points, with a being the one inside σ. Then $w(\sigma') = \overline{ab} + \overline{bc} \leq w + \overline{bc}$.

We have $\overline{bc} = \overline{oc} \cdot \tan(\alpha - \gamma) \leq \overline{oc} \cdot \tan(\delta\theta)$. We consider the following two cases.

Case 1. $\widetilde{w}/D \leq 1/2$**.** That is, $\theta \leq \pi/6$. In this case, since $\delta < 1$ and $\theta \leq \pi/6$, we have $\tan(\delta\theta) \leq \delta \tan\theta$. Note that $\delta\tan\theta = \delta\sin\theta/\cos\theta \leq (2/\sqrt{3})\delta\sin\theta \leq (2/\sqrt{3})\delta\widetilde{w}/D \leq (2/\sqrt{3})c\delta w/D$. Since $\overline{oc} \leq diam(P) \leq 4D$, it follows that $\overline{bc} \leq (8/\sqrt{3})c\delta w$.

Case 2. $\widetilde{w}/D > 1/2$**.** We have $\tan(\delta\theta) = \sin(\delta\theta)/\cos(\delta\theta) \leq 2 \cdot \sin(\delta\theta) \leq 2\delta\theta \leq \pi\delta$ since $\delta\theta \leq \pi/3$ assuming $\delta \leq 2/3$. Also, $\overline{oc} \leq diam(P) \leq 4D \leq 8\widetilde{w} \leq 8cw$. Combining these bounds gives $\overline{bc} \leq 8\pi\delta cw$.

By choosing $\delta = \min\{2/3, \sqrt{3}\varepsilon/(8c), \varepsilon/(8\pi c)\}$, it holds that $\overline{bc} \leq \varepsilon w$ in both cases. □

By Lemma 2, we can compute a set F of at most 11 pairs such that one

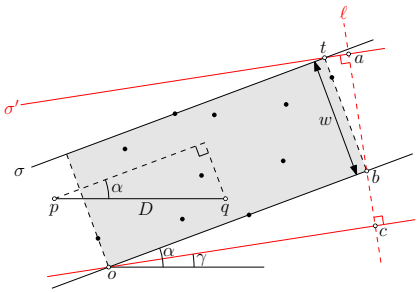


Figure 18: The black points represent the points in P and the gray region represents the minimum bounding box of P with orientation α.

of them is an anchor pair of Σ^* in $O(n)$ time. For the moment, assume that we have an anchor pair (p, q) of Σ^*. (We will repeat the following process for every pair in F.) Without loss of generality, assume that $p, q \in \sigma_1^*$ and $d(p, q) \geq diam((\sigma_1^* \setminus \sigma_2^*) \cap S)/4$.

By Theorem 5, we can compute $\widetilde{w}$ such that $w^* \leq \widetilde{w} \leq 10w^*$ in $O(n)$ time.

Consider a slab σ_0^* that has the same orientation as σ_1^*, has width w^*, and contains σ_1^*. In particular, if $w(\sigma_1^*) = w^*$, then σ_0^* is σ_1^*.

Observe that $P = S \cap (\sigma_1^* \setminus \sigma_2^*)$ and σ_0^* satisfy the four conditions in the statement of Lemma 20 for (p, q) and the value $\widetilde{w}$ computed above, with $c = 10$. Therefore, in $O(1/\varepsilon)$ time, we can compute a set Γ of $O(1/\varepsilon)$ orientations such that there exists a slab σ' with orientation $\gamma \in \Gamma$ that covers $S \cap (\sigma_1^* \setminus \sigma_2^*)$ and satisfies $w(\sigma') \leq (1 + \varepsilon)w^*$. This implies that there exists a pair of slabs whose union covers S, one of which has an orientation from Γ, and whose maximum width is at most $(1 + \varepsilon)w^*$.

For every $\gamma \in \Gamma$, we find a pair of slabs whose union covers S, one of which has orientation γ, and whose maximum width is minimized. By Theorem 18, this can be done in $O((n/\varepsilon) \log n)$ time in total. We repeat this procedure for every pair in F, which adds only a constant factor to the running time. Observe that each of the pairs of slabs we obtain has the property that its union covers S. Among these pairs, we output the one with the smallest maximum width. Thus, we conclude the following lemma.

Lemma 21. *We can find a pair of slabs such that their union covers S and the maximum of their widths is at most $(1 + \varepsilon)w^*$ in $O((n/\varepsilon) \log n)$ time.*

To further refine the logarithmic term, we compute an ε-certificate Q of S (see Section 3.3). By Theorem 8, this can be done in $O(n)$ time and $|Q| = O(1/\varepsilon^2)$. We then compute a $(1 + \varepsilon)$-approximate pair (σ_1, σ_2) for the general two-line-center problem on Q, which can be done in $O((|Q|/\varepsilon) \log |Q|)$ time by Lemma 21. Without loss of generality, we assume that σ_1 and σ_2 have the same width (otherwise, we expand the smaller one). Let $w = w(\sigma_1) = w(\sigma_2)$. Since $Q \subseteq S$, it is clear that $w \leq (1 + \varepsilon)w^*$.

Let (σ_1', σ_2') be the pair of slabs obtained by taking the ε-expansions of σ_1 and σ_2, respectively. By definition, $\sigma_1' \cup \sigma_2'$ covers S, and $w(\sigma_1') = w(\sigma_2') \leq (1 + \varepsilon)^2 w^* \leq (1 + 3\varepsilon)w^*$.

The total running time is $O(n + (\mu/\varepsilon) \log \mu)$, where $\mu = \min\{n, 1/\varepsilon^2\}$, which can also be expressed as both $O((n/\varepsilon) \log(1/\varepsilon))$ and $O(n + (1/\varepsilon^3) \log(1/\varepsilon))$. By

replacing ε with $\varepsilon/3$, we obtain the following theorem.

Theorem 22. *Given a set S of n points in the plane, we can find a pair of slabs such that their union covers S and the maximum of their widths is at most $(1+\varepsilon)w^*$ in $O(n + (\mu/\varepsilon)\log\mu)$ time, where $\mu = \min\{n, 1/\varepsilon^2\}$.*

6 Two Fixed Orientations

By rotating the coordinate system, we assume without loss of generality that one of the given orientations is horizontal. Thus, the two orientations can be taken as 0 and θ, where $0 \le \theta < \pi$. Let (σ_h^*, σ_t^*) be an optimal pair of slabs where σ_h^* is horizontal and σ_t^* has orientation θ. Let $w^* = \max\{w(\sigma_h^*), w(\sigma_t^*)\}$. Recall that S denote the input set of n points in the plane.

6.1 A 2-Approximation Algorithm

Let p_t and p_b denote the highest and lowest points in S, respectively. Let p_ℓ and p_r be the two extreme points in S in the direction orthogonal to θ. See Figure 19. Since we assume $w^* > 0$, the size of $\{p_b, p_t, p_\ell, p_r\}$ is at least two. Then the following holds.

Lemma 23. *There exists an optimal pair (σ_h^*, σ_t^*) of slabs such that two distinct points p and q from $\{p_b, p_t, p_\ell, p_r\}$ satisfy $p \in \sigma_h^*$ and $q \in \sigma_t^*$. (Note that p or q may belong to $\sigma_h^* \cap \sigma_t^*$.)*

Proof. Suppose to the contrary. Then for every optimal pair (σ_h^*, σ_t^*) of slabs, either $\{p_b, p_t, p_\ell, p_r\} \subset \sigma_h^* \setminus \sigma_t^*$ or $\{p_b, p_t, p_\ell, p_r\} \subset \sigma_t^* \setminus \sigma_h^*$. By definition, the points $\{p_b, p_t, p_\ell, p_r\}$ determine the minimum bounding parallelogram of S; two of its sides are horizontal and the other two sides have orientation θ (see Figure 19). Let R denote the parallelogram.

If $\{p_b, p_t, p_\ell, p_r\} \subset \sigma_h^* \setminus \sigma_t^*$, then $R \subset \sigma_h^*$, and hence $S \subset \sigma_h^*$. In this case, choose any point $p \in \{p_b, p_t, p_\ell, p_r\}$ and consider the line ℓ_p of orientation θ passing through p. Then (σ_h^*, ℓ_p) forms an optimal pair of slabs, with one slab degenerating to a line. This contradicts our assumption, since any point in $\{p_b, p_t, p_\ell, p_r\} \setminus \{p\}$ lies in σ_h^* while $p \in \ell_p$.

Analogously, if $\{p_b, p_t, p_\ell, p_r\} \subset \sigma_t^* \setminus \sigma_h^*$, consider the horizontal line ℓ_p passing through $p \in \{p_b, p_t, p_\ell, p_r\}$. Then (ℓ_p, σ_t^*) forms an optimal pair of slabs, which contradicts our assumption in the same way. □

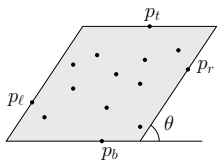


Figure 19: The four extreme points $\{p_b, p_t, p_\ell, p_r\}$ determine the minimum bounding parallelogram.

Using Lemma 23, we first give a 2-approximation algorithm.

Lemma 24. *We can compute a 2-approximation of an optimal pair of slabs in $O(n)$ time.*

Proof. For the moment, assume that we are given a pair (p_1, p_2) of points in S such that $p_1 \in \sigma_h^*$ and $p_2 \in \sigma_t^*$. Let ℓ_h be the horizontal line through p_1, and let ℓ_t be the line of orientation θ through p_2. Then we have $\ell_h \subset \sigma_h^*$ and $\ell_t \subset \sigma_t^*$.

Now we determine by which slab each input point should be covered in a greedy way. We initialize two sets H and T as empty sets. For each $p \in S$, we insert p into H if $d(p, \ell_h) < d(p, \ell_t)$. Otherwise, we insert it into T. After applying this process to every input point, we compute the minimum horizontal slab enclosing H, denoted by σ_h, and the minimum slab of orientation θ enclosing T, denoted by σ_t (see Figure 20). Now we show that (σ_h, σ_t) is a 2-approximate solution.

We first show that $w(\sigma_h) \leq 2w^*$. Let q and q' be the lowest and highest points in H (see Figure 20). Then $w(\sigma_h) \leq d(q, \ell_h) + d(q', \ell_h)$. (Equality holds if q and q' are on different sides of ℓ_h.) If $q \in \sigma_h^*$, then $d(q, \ell_h) \leq w^*$ as $\ell_h \subset \sigma_h^*$. Otherwise, $d(q, \ell_t) \leq w^*$ as q should be covered by σ_t^*. We also have $d(q, \ell_h) < d(q, \ell_t)$ as $q \in H$. Therefore it holds that $d(q, \ell_h) \leq w^*$ in any case. Similarly, we can also show that $d(q', \ell_h) \leq w^*$. Therefore, $w(\sigma_h) \leq d(q, \ell_h) + d(q', \ell_h) \leq 2w^*$. We can similarly show that $w(\sigma_t) \leq 2w^*$ by choosing the two extreme points of T in the direction orthogonal to θ. Note that once we have p_1 and p_2, we can find σ_h and σ_t in $O(n)$ time by finding the extreme points in H and T, respectively.

By Lemma 23, we have a constant number of candidates for (p_1, p_2). We apply the above procedure to each candidate pair and select the one that minimizes $\max\{w(\sigma_h), w(\sigma_t)\}$. The resulting pair (σ_h, σ_t) is a 2-approximate solution. □

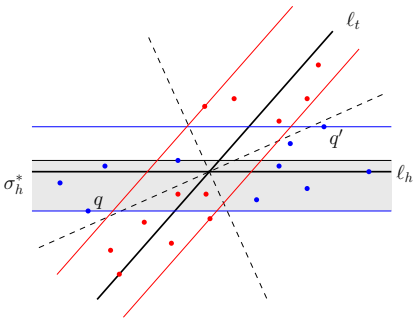


Figure 20: Two lines ℓ_h and ℓ_t and their bisectors (dashed). Blue points belong to H, while red points belong to T. The blue lines indicate the bounding lines of σ_h, and the red lines indicate the bounding lines of σ_t. Points q and q' are the vertical extreme points of H. The gray region represents σ_h^*. Note that $q \in \sigma_h^*$ and $q' \notin \sigma_h^*$.

6.2 A $(1+\varepsilon)$-Approximation Algorithm

By Lemma 24, we can compute $\widetilde{w}$ such that $w^* \leq \widetilde{w} \leq 2w^*$ in $O(n)$ time. By Lemma 23, we can compute a set G of at most six pairs of points such that there exists a pair $(p_1, p_2) \in G$ where $p_1 \in \sigma_h^*$ and $p_2 \in \sigma_t^*$ in $O(n)$ time. For

the moment, assume that we have such a pair (p_1, p_2) of G. (We will repeat the following process for every pair in G.)

Let ℓ_h be the horizontal line through p_1, and let ℓ_t be the line of orientation θ through p_2. Then we have $\ell_h \subset \sigma_h^*$ and $\ell_t \subset \sigma_t^*$. Without loss of generality, we assume that ℓ_h is $y = 0$. We also assume that $w(\sigma_h^*) \geq w(\sigma_t^*)$. We can handle the other case by running the same algorithm in the coordinate system rotated by θ.

Let $L = \{\ell_i : y = (\varepsilon\widetilde{w}/4) \cdot (i - 1 - \lceil 4/\varepsilon \rceil) \mid 1 \leq i \leq 2\lceil 4/\varepsilon \rceil + 1\}$ be the set of horizontal lines. Note that ℓ_1 is $y = -(\varepsilon\widetilde{w}/4) \cdot \lceil 4/\varepsilon \rceil \leq -\widetilde{w}$ and $\ell_{2\lceil 4/\varepsilon \rceil + 1}$ is $y = (\varepsilon\widetilde{w}/4) \cdot \lceil 4/\varepsilon \rceil \geq \widetilde{w}$. The distance between two consecutive lines is $\varepsilon\widetilde{w}/4$ and $|L| = 2\lceil 4/\varepsilon \rceil + 1$.

For two indices $1 \leq i \leq j \leq 2\lceil 4/\varepsilon \rceil + 1$, let $\sigma(i, j)$ be the horizontal slab bounded by ℓ_i and ℓ_j (see Figure 21). If $i = j$, then $\sigma(i, j)$ is ℓ_i, which covers the points lying on it.

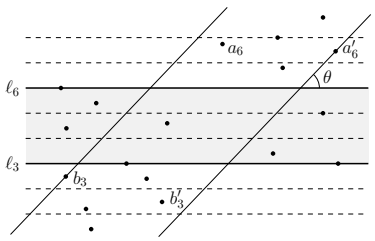


Figure 21: The lines in L (dashed), ℓ_3, and ℓ_6 are shown. The gray region represents $\sigma(3, 6)$. Points a_6 and a'_6 (resp., b_3 and b'_3) are the two extreme points in the direction orthogonal to θ among the points lying strictly above ℓ_6 (resp., strictly below ℓ_3). Note that the minimum slab of orientation θ enclosing the points outside $\sigma(3, 6)$ is determined by a'_6 and b_3.

Lemma 25. *There is a horizontal slab σ such that both bounding lines are in L, $\sigma_h^* \subseteq \sigma$, and $w(\sigma) \leq (1 + \varepsilon)w^*$.*

Proof. Let i' and j' be the indices such that $\sigma(i', j')$ is the minimal slab that encloses σ_h^* among all slabs $\sigma(i, j)$ with $1 \leq i \leq j \leq 2\lceil 4/\varepsilon \rceil + 1$. Such indices exist because $d(\ell_1, \ell_h) \geq \widetilde{w} \geq w^*$ and $d(\ell_{2\lceil 4/\varepsilon \rceil + 1}, \ell_h) \geq \widetilde{w} \geq w^*$, and $\ell_h \subset \sigma_h^*$. Then $\sigma_h^* \subseteq \sigma(i', j')$ and $w(\sigma(i', j')) \leq w(\sigma_h^*) + \varepsilon\widetilde{w}/2$. Since $w(\sigma_h^*) \leq w^*$ and $\widetilde{w} \leq 2w^*$, it follows that $w(\sigma(i', j')) \leq (1 + \varepsilon)w^*$. □

Observe that once we find a slab σ satisfying $\sigma_h^* \subseteq \sigma$ and $w(\sigma) \leq (1 + \varepsilon)w^*$, the other slab is determined immediately: it is the minimum-width slab σ' of orientation θ that encloses all points not covered by σ. Note that (σ, σ') forms a $(1 + \varepsilon)$-approximate solution.

By Lemma 25, there are at most $|L|^2 = O(1/\varepsilon^2)$ candidates for such a slab σ. However, we preprocess the input points so that we avoid testing all $O(1/\varepsilon^2)$ candidates.

Preprocessing The lines in L partition the plane into $O(1/\varepsilon)$ regions. For each point $p \in S$, we determine the region containing p, which can be done in constant time using the floor function. Within each region, we find the two extreme points in the direction orthogonal to θ. This takes $O(n)$ time in total.

For each line $\ell_i \in L$, we store the two extreme points in the direction orthogonal to θ for the points that lie strictly above ℓ_i, and denote them by a_i and a'_i (see Figure 21). Similarly, for each line $\ell_i \in L$, we store the two extreme points for those lying strictly below ℓ_i, and denote them by b_i and b'_i. This can be done in $O(n)$ total time by visiting the partitioned regions in order.

Algorithm For fixed i and j, $1 \le i \le j \le 2\lceil 4/\varepsilon \rceil + 1$, note that the smallest slab of orientation θ that encloses the input points strictly outside of $\sigma(i,j)$ is the smallest slab of orientation θ that encloses $\{a_i, a'_i, b_j, b'_j\}$ (see Figure 21). That is, $\mathrm{width}_\theta(S \setminus \sigma(i,j)) = \mathrm{width}_\theta(\{a_i, a'_i, b_j, b'_j\})$. (Recall the definition of width_θ in Section 2.) We can compute $\mathrm{width}_\theta(S \setminus \sigma(i,j))$ in constant time once we know $\{a_i, a'_i, b_j, b'_j\}$.

For $1 \le i, j \le 2\lceil 4/\varepsilon \rceil + 1$, we define $M[i,j] = d(\ell_i, \ell_j)$ if $i \le j$ and $d(\ell_i, \ell_j) \ge w_\theta(S \setminus \sigma(i,j))$. Let $M[i,j] = -\infty$ if $i > j$ or $d(\ell_i, \ell_j) < w_\theta(S \setminus \sigma(i,j))$. We first show that M is a sorted matrix in the following lemma. A smallest non-negative element in a sorted matrix can be found in time that is linear in the number of rows plus the number of columns if each element can be computed in constant time. Observe that each element of M can be computed in constant time by the preprocessing step.

Lemma 26. *M is a sorted matrix. In particular, each row is a non-decreasing sequence and each column is a non-increasing sequence.*

Proof. Let $t = 2\lceil 4/\varepsilon \rceil + 1$, which is the number of rows (resp., the number of columns) of M.

Fix an index i with $1 \le i \le t$. For every j with $1 \le j < i$, we have $M[i,j] = -\infty$. Let k be the smallest index with $i \le k \le t$ such that $d(\ell_i, \ell_k) \ge w_\theta(S \setminus \sigma(i,k))$. If no such k exists, then $M[i,j] = -\infty$ for all $j \ge i$. If such a k exists, then $M[i,j] = -\infty$ for all j with $i \le j < k$, and $M[i,j] = d(\ell_i, \ell_j)$ for all j with $k \le j \le t$. In both cases, the i-th row of M is a non-decreasing sequence.

Fix an index j with $1 \le j \le t$. For every i with $j < i$, we have $M[i,j] = -\infty$. Let k' be the largest index with $1 \le k' \le j$ such that $d(\ell_{k'}, \ell_j) \ge w_\theta(S \setminus \sigma(k', j))$. If no such k' exists, then $M[i,j] = -\infty$ for all $i \le j$. If such a k' exists, then $M[i,j] = -\infty$ for all i with $k' < i \le j$, and $M[i,j] = d(\ell_i, \ell_j)$ for all i with $1 \le i \le k'$. In both cases, the j-th column of M is a non-increasing sequence. □

Let (i', j') be the pair of indices such that $\sigma(i', j')$ is the minimal slab that encloses σ_h^* among all slabs $\sigma(i,j)$ with $1 \le i \le j \le 2\lceil 4/\varepsilon \rceil + 1$. By our assumption that $w(\sigma_h^*) \ge w(\sigma_t^*)$, we have $d(\ell_{i'}, \ell_{j'}) \ge w(\sigma_h^*) \ge w(\sigma_t^*) \ge w_\theta(S \setminus \sigma(i', j'))$. By Lemma 25, $d(\ell_{i'}, \ell_{j'}) \le (1+\varepsilon)w^*$. Therefore, $w^* \le M[i', j'] = d(\ell_{i'}, \ell_{j'}) \le (1+\varepsilon)w^*$, which implies that a smallest non-negative element in M exists and is at most $(1+\varepsilon)w^*$.

We compute a pair (i'', j'') of indices such that $M[i'', j'']$ is a smallest non-negative element in M. This can be done in $O(1/\varepsilon)$ time. This directly yields a pair of slabs: $\sigma(i'', j'')$ and the smallest slab of orientation θ that encloses $S \setminus \sigma(i'', j'')$. Observe that these two slabs together form a $(1+\varepsilon)$-approximate pair of slabs.

Now, we conclude the following theorem.

Theorem 27. *Given a set S of n points in the plane and two orientations θ_1 and θ_2, we can find a pair of slabs, one with orientation θ_1 and the other with*

orientation θ_2, such that their union covers S and their maximum width is at most $(1+\varepsilon)w^*$ in $O(n+1/\varepsilon)$ time.

7 Parallel Two-Line-Center

Throughout this section, let $\Sigma^* = (\sigma_1^*, \sigma_2^*)$ be an optimal pair of slabs for the parallel-two-line-center problem. Let $w^* = \max\{w(\sigma_1^*), w(\sigma_2^*)\}$. Recall that S denotes the input set of n points in the plane.

Notation Let ρ be a fixed constant with $1/2 \leq \rho < 1$. Given a pair of interior-disjoint parallel slabs $\Sigma = (\sigma_1, \sigma_2)$, let $W(\Sigma)$ denote the largest distance between a boundary of σ_1 and a boundary of σ_2. Let $g(\Sigma)$ denote the smallest distance between a boundary of σ_1 and a boundary of σ_2. See Figure 22.

Definition 28 (Gap-ratio)**.** *Given a pair of interior-disjoint parallel slabs $\Sigma = (\sigma_1, \sigma_2)$, the* gap-ratio *of Σ is defined as $g(\Sigma)/W(\Sigma)$. If $W(\Sigma) = 0$, the gap-ratio is not defined.*

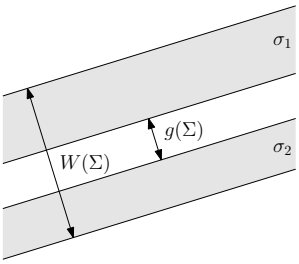


Figure 22: For a pair of interior-disjoint parallel slabs $\Sigma = (\sigma_1, \sigma_2)$, we denote by $W(\Sigma)$ and $g(\Sigma)$ the largest and smallest distances, respectively, between a boundary of σ_1 and a boundary of σ_2.

To handle this problem, we consider two cases: whether the gap-ratio of Σ^* is at most ρ, or not. In the following, we present two different algorithms, one for each case. In Sections 7.1 and 7.2, we prove the corresponding theorems.

Theorem 29 (Small gap-ratio)**.** *Assuming that there exists an optimal pair $\Sigma^* = (\sigma_1^*, \sigma_2^*)$ for the parallel two-line-center problem whose gap ratio is at most ρ, we can compute a pair of parallel slabs such that their union covers S and the maximum of their widths is at most $(1+\varepsilon)w^*$ in $O(n/\varepsilon)$ time.*

Theorem 30 (Large gap-ratio)**.** *Assuming that there exists an optimal pair $\Sigma^* = (\sigma_1^*, \sigma_2^*)$ for the parallel two-line-center problem whose gap ratio is at least ρ, we can compute a pair of parallel slabs such that their union covers S and the maximum of their widths is at most $(1+\varepsilon)w^*$ in $O(n + \mu \log \mu)$ time, where $\mu = \min\{n, 1/\varepsilon^2\}$. If the assumption on the gap-ratio is not satisfied, the algorithm may return NO.*

We first prove Theorem 31 by combining Theorem 29 and Theorem 30 with an ε-certificate (see Section 3.3).

Theorem 31. *Given a set S of n points in the plane, we can find a pair of parallel slabs such that their union covers S and the maximum of their widths is at most $(1+\varepsilon)w^*$ in $O(n + \mu/\varepsilon)$ time, where $\mu = \min\{n, 1/\varepsilon^2\}$.*

Proof. We first compute an ε-certificate $Q \subseteq S$ of S. By Theorem 8, this takes $O(n)$ time, and $|Q| = O(\mu)$ where $\mu = \min\{n, 1/\varepsilon^2\}$. Let $w^*(Q)$ denote the maximum width of an optimal solution for the parallel two-line-center problem on Q. It is clear that $w^*(Q) \leq w^*$ since $Q \subseteq S$.

We run the algorithms of Theorem 29 and Theorem 30 on Q. Observe that at least one of them returns a pair of parallel slabs whose union covers Q and whose maximum width is at most $(1+\varepsilon)w^*(Q)$. Among the (at most) two pairs, let (σ_1, σ_2) be the one whose maximum width is smaller. Without loss of generality, assume that σ_1 and σ_2 have the same width (otherwise, we expand the smaller one). Therefore, $w(\sigma_1) = w(\sigma_2) \leq (1+\varepsilon)w^*(Q)$.

Let (σ_1', σ_2') be the pair of slabs obtained by taking the ε-expansions of σ_1 and σ_2, respectively. By the definition of an ε-expansion, we have $w(\sigma_1') \leq (1+\varepsilon)w(\sigma_1) \leq (1+\varepsilon)^2 w^*(Q) \leq (1+\varepsilon)^2 w^*$. Therefore, $w(\sigma_1') \leq (1+O(\varepsilon))w^*$, and the same bound holds for $w(\sigma_2')$. Moreover, by the definition of an ε-certificate, $\sigma_1' \cup \sigma_2'$ covers S. Therefore, (σ_1', σ_2') forms a $(1+O(\varepsilon))$-approximate solution.

The total running time is $O(n + \mu/\varepsilon + \mu \log \mu)$ time, which is $O(n + \mu/\varepsilon)$. This can be expressed as both $O(n/\varepsilon)$ and $O(n + 1/\varepsilon^3)$. □

7.1 Small Gap-Ratio (proof of Theorem 29)

In this section, we assume that there exists an optimal pair $\Sigma^* = (\sigma_1^*, \sigma_2^*)$ for the parallel two-line-center problem whose gap ratio is at most ρ. Recall that $w^* = \max\{w(\sigma_1^*), w(\sigma_2^*)\}$. We present an $O(n/\varepsilon)$-time algorithm that outputs a pair of parallel slabs such that their union covers S and the maximum of their widths is at most $(1+\varepsilon)w^*$ under the assumption.

Observation 32. *For a pair $\Sigma = (\sigma_1, \sigma_2)$ of parallel slabs whose gap-ratio is at most ρ, it holds that $g(\Sigma) \leq \frac{2\rho}{1-\rho} \cdot w_{\max}$ where $w_{\max} = \max\{w(\sigma_1), w(\sigma_2)\}$.*

Proof. Let $g = g(\Sigma)$ and $W = W(\Sigma)$. By definition, we have $W = w(\sigma_1) + w(\sigma_2) + g \leq 2w_{\max} + g$. Since the gap-ratio of Σ is at most ρ, it holds that $\rho \geq g/W \geq g/(2w_{\max} + g)$. Therefore, $2\rho \cdot w_{\max} \geq (1-\rho)g$, which implies $g \leq \frac{2\rho}{1-\rho} \cdot w_{\max}$. □

Lemma 33. *We can compute $\widetilde{w}$ such that $w^* \leq \widetilde{w} \leq (2 + \frac{2\rho}{1-\rho})w^*$ in $O(n)$ time.*

Proof. We first compute w' such that $\mathrm{width}(S) \leq w' \leq 2 \cdot \mathrm{width}(S)$ using a 2-approximation algorithm for the width, which can be done in $O(n)$ time [12]. Let $\widetilde{w} = w'/2$. We show that $\widetilde{w}$ satisfies the condition in the statement. Observe that $w^* \leq \mathrm{width}(S)/2$ as a slab of width(S) enclosing S directly gives us a pair of parallel slabs of width width$(S)/2$ whose union covers S. Therefore, we have $w^* \leq \mathrm{width}(S)/2 \leq \widetilde{w}$. It holds that $\mathrm{width}(S) \leq W(\Sigma^*) = g(\Sigma^*) + w(\sigma_1^*) + w(\sigma_2^*) \leq g(\Sigma^*) + 2w^*$. Since $\widetilde{w} \leq \mathrm{width}(S)$ and by Observation 32, we obtain $\widetilde{w} \leq (2 + \frac{2\rho}{1-\rho})w^*$. □

Lemma 34. *Let p and q be two points in $\mathbb{R}^2$ and let $\widetilde{w} > 0$ and $\varepsilon > 0$ be real numbers. In $O(1/\varepsilon)$ time, we can compute a set Γ of $O(1/\varepsilon)$ orientations, such that the following holds:*

For every finite set P of points in $\mathbb{R}^2$ and for every pair of parallel slabs $\Sigma = (\sigma_1, \sigma_2)$, if

(1) The gap-ratio of Σ *is at most* ρ*,*

(2) $P \subset \sigma_1 \cup \sigma_2$*,*

(3) $w_{\max} \leq \widetilde{w} \leq c \cdot w_{\max}$ *where* $w_{\max} = \max\{w(\sigma_1), w(\sigma_2)\}$ *and* $c = 2 + \frac{2\rho}{1-\rho}$*,*

(4) $p, q \in \sigma_1 \cup \sigma_2$ *and* $d(p,q) \geq \mathrm{diam}(P)/2$*,*

then there exists an orientation $\gamma \in \Gamma$ *and a pair of parallel slabs* $\Sigma' = (\sigma'_1, \sigma'_2)$*, such that*

(1) $P \subset \sigma'_1 \cup \sigma'_2$*,*

(2) $w'_{\max} \leq (1+\varepsilon) \cdot w_{\max}$ *where* $w'_{\max} = \max\{w(\sigma'_1), w(\sigma'_2)\}$*,*

(3) both σ'_1 *and* σ'_2 *have the orientation* γ*.*

Proof. We remark that the idea of this proof is based on the approach in [2]. Without loss of generality, assume that p and q lie on a horizontal line. Let $D = d(p,q)$. Let θ be the orientation with $0 \leq \theta \leq \pi/2$ such that $\sin\theta = \min\{1, c\widetilde{w}/D\}$ for $c = 2 + \frac{2\rho}{1-\rho}$. Let $\delta = c_\delta \varepsilon$ for some constant $0 < c_\delta < 1$. Let $\Gamma = \{\gamma_i = (i - \lceil \frac{1}{\delta} \rceil) \cdot \delta\theta \mid 0 \leq i \leq 2\lceil \frac{1}{\delta} \rceil\}$. Observe that Γ contains uniformly placed orientations in the range $[-\theta, \theta]$, and $|\Gamma| = O(1/\varepsilon)$. (Here we slightly abuse the notation by redefining orientations to lie in $[-\pi/2, \pi/2]$ instead of $[0, \pi]$.) In the following, we prove that Γ satisfies the conditions in the statement.

Let P be set of points in $\mathbb{R}^2$ and $\Sigma = (\sigma_1, \sigma_2)$ be a pair of parallel slabs, where P and Σ satisfy the four conditions in the statement. Let α be the orientation of the slabs in Σ. For simplicity, assume that $0 \leq \alpha \leq \pi/2$. The other case can be shown in a symmetric way. Observe that $\sin\alpha \leq W(\Sigma)/D$, even when $\alpha = \pi/2$ (see the right triangle having $\overline{pq}$ as one of its sides in Figure 23). We have $W(\Sigma) = g(\Sigma) + w(\sigma_1) + w(\sigma_2) \leq g(\Sigma) + 2w_{\max}$. By Observation 32 and as $w_{\max} \leq \widetilde{w}$, $W(\Sigma) \leq (2 + \frac{2\rho}{1-\rho}) w_{\max} \leq c\widetilde{w}$. Therefore, we have $\sin\alpha \leq c\widetilde{w}/D$. Observe that if $c\widetilde{w}/D < 1$, then $\sin\theta = c\widetilde{w}/D$. If $c\widetilde{w}/D \geq 1$, then $\sin\theta = 1$. In both cases, it holds that $\sin\alpha \leq \sin\theta$. Therefore, we have $\alpha \leq \theta$.

Let γ be the largest orientation in Γ such that $\gamma \leq \alpha$. Since $\alpha \leq \theta$, we have $\alpha - \gamma < \delta\theta$. Let R_1 be the minimum bounding box of $P \cap \sigma_1$ with orientation α. See Figure 23. Denote by o and t the lowest and highest vertices of R_1, respectively. Now consider a slab σ'_1 of orientation γ whose two bounding lines pass through o and t. Clearly, $P \cap \sigma_1 \subset \sigma'_1$. We next show that $w(\sigma'_1) \leq (1+\varepsilon) w_{\max}$.

Let b be a vertex of R_1 other than o and t. Consider the line ℓ orthogonal to σ'_1 that passes through b. The line ℓ intersects the two bounding lines of σ'_1. Observe that one intersection point lies inside σ_1, while the other lies outside. Let a and c denote these two intersection points, with a being the one inside σ_1. Then $w(\sigma'_1) = \overline{ab} + \overline{bc} \leq w(\sigma_1) + \overline{bc} \leq w_{\max} + \overline{bc}$.

We have $\overline{bc} = \overline{oc} \cdot \tan(\alpha - \gamma) \leq \overline{oc} \cdot \tan(\delta\theta)$. Observe that $\overline{oc} \leq \overline{ob} \leq \mathrm{diam}(P)$. As $\mathrm{diam}(P)/2 \leq D$, we have $\overline{oc} \leq 2D$. We now consider the following two cases.

Case 1. $c\widetilde{w}/D \leq 1/2$**.** That is, $\theta \leq \pi/6$. In this case, since $\delta < 1$ and $\theta \leq \pi/6$, we have $\tan(\delta\theta) \leq \delta \tan\theta$. Note that $\delta \tan\theta = \delta \sin\theta / \cos\theta \leq (2/\sqrt{3}) \delta \sin\theta = (2/\sqrt{3}) c\delta\widetilde{w}/D \leq (2/\sqrt{3}) c^2 \delta w_{\max}/D$. Since $\overline{oc} \leq 2D$, it follows that $\overline{bc} \leq (4/\sqrt{3}) c^2 \delta w_{\max}$.

Case 2. $c\widetilde{w}/D > 1/2$. Since $\delta\theta \le \pi/3$ (because $\theta \le \pi/2$ and we assume $\delta \le 2/3$), we have $\tan(\delta\theta) = \sin(\delta\theta)/\cos(\delta\theta) \le 2 \cdot \sin(\delta\theta) \le 2\delta\theta \le \pi\delta$. Also, $\overline{oc} \le 2D \le 4c\widetilde{w} \le 4c^2 w_{\max}$. Combining these bounds gives $\overline{bc} \le 4\pi c^2 \delta w_{\max}$.

By choosing $\delta = \min\{2/3, \sqrt{3}\varepsilon/(4c^2), \varepsilon/4\pi c^2\}$, it holds that $\overline{bc} \le \varepsilon w_{\max}$ in both cases. Therefore, we have $w(\sigma_1') \le (1+\varepsilon)w_{\max}$.

In a symmetric way, we obtain σ_2' of orientation γ and it can be shown that $P \cap \sigma_2 \subset \sigma_2'$ and $w(\sigma_2') \le (1+\varepsilon)w_{\max}$. □

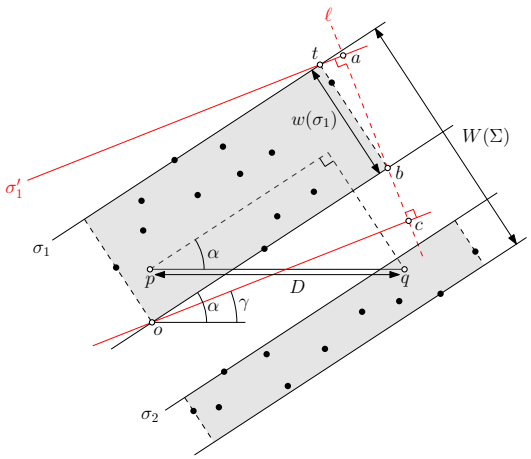


Figure 23: A pair (σ_1, σ_2) of parallel slabs with orientation α and two points $p, q \in \sigma_1 \cup \sigma_2$ are shown. The black points represent the points in P. The two gray regions represent the minimum bounding boxes of $P \cap \sigma_1$ and $P \cap \sigma_2$ with orientation α, respectively.

Observation 35. *Given an orientation θ, we can find a pair of parallel slabs of orientation θ such that their union covers S and their maximum width is minimized in $O(n)$ time.*

Proof. We can compute the minimum-width slab σ of orientation θ that encloses S in $O(n)$ time by finding two extreme points in the direction orthogonal to θ. Let ℓ_1 and ℓ_2 be the two bounding lines of σ, and let m be its center line. See Figure 24.

Let m_1 be the line obtained by translating m toward ℓ_1 until it touches the first point of S. Symmetrically, let m_2 be the line obtained by translating m toward ℓ_2 until it touches the first point of S. Both m_1 and m_2 can be computed in $O(n)$ time.

Let σ_1 be the slab having ℓ_1 and m_1 as its bounding lines, and let σ_2 be the slab having ℓ_2 and m_2 as its bounding lines. Observe that (σ_1, σ_2) is a pair of slabs of orientation θ whose union encloses S while minimizing the maximum of their widths. □

Now we prove Theorem 29 using the above observations and lemmas. We first compute a pair of points p, q in S where $d(p,q) \ge \mathrm{diam}(S)/2$, which can be done in $O(n)$ time. By Lemma 33, we can compute $\widetilde{w}$ with $w^* \le \widetilde{w} \le (2+\frac{2\rho}{1-\rho})w^*$ in $O(n)$ time.

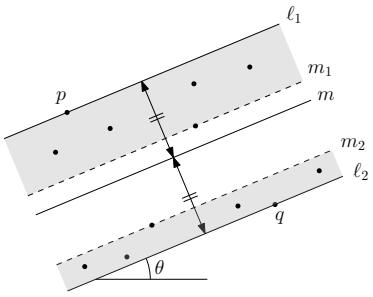


Figure 24: Black points indicate the point in S. Observe that p and q are the extreme points in the direction orthogonal to θ. The lines ℓ_1, ℓ_2, m_1, m_2, and m are shown as described above. Observe that the pair (σ_1, σ_2), represented by the two gray regions in the figure, is a pair of slabs of orientation θ whose union encloses S while minimizing the maximum of their widths.

Observe that S and Σ^* satisfy the four conditions in the statement of Lemma 34 for (p, q) and the value $\widetilde{w}$ computed above. Therefore, in $O(1/\varepsilon)$ time, we can compute a set Γ of $O(1/\varepsilon)$ orientations, such that there exists a pair of parallel slabs $\Sigma' = (\sigma'_1, \sigma'_2)$ of orientation $\gamma \in \Gamma$ where $S \subset \sigma'_1 \cup \sigma'_2$ and $\max\{w(\sigma'_1), w(\sigma'_2)\} \leq (1+\varepsilon)w^*$.

Therefore, for every orientation $\gamma \in \Gamma$, we compute a pair of parallel slabs of orientation γ whose union covers S while minimizing the maximum of their widths. Among these $O(1/\varepsilon)$ pairs of slabs, we output the one that minimizes the maximum of their widths. By Observation 35, it takes $O(n/\varepsilon)$ time in total.

7.2 Large Gap-Ratio (proof of Theorem 30)

In this section, we assume that there exists an optimal pair $\Sigma^* = (\sigma_1^*, \sigma_2^*)$ for the parallel two-line-center problem whose gap ratio is at least ρ. Recall that $w^* = \max\{w(\sigma_1^*), w(\sigma_2^*)\}$. We present an algorithm that outputs a pair of parallel slabs such that their union covers S and the maximum of their widths is at most $(1+\varepsilon)w^*$ under the assumption. It runs in $O(n + \mu \log \mu)$ time, where $\mu = \min\{n, 1/\varepsilon^2\}$, which can be expressed as both $O(n \log(1/\varepsilon))$ and $O(n + (1/\varepsilon^2)\log(1/\varepsilon))$.

We first recall an algorithm of [13] that computes a pair of parallel slabs with a gap-ratio at least ρ whose union cover S while minimizing their maximum width.

Theorem 36 ([13])**.** *Given a set P of n points in the plane and a constant $0 < \rho < 1$, we can do the following in $O(n \log n)$ time. If there exists a pair of parallel slabs such that (1) their union covers P, (2) each slab contains at least one point in P, and (3) its gap-ratio is at least ρ, then we report one such pair that minimizes the maximum of their widths. Otherwise, we report NO.*

By our assumption that $w^* > 0$ (see Appendix D for the case where $w^* = 0$), we have $S \cap \sigma_1^* \neq \emptyset$ and $S \cap \sigma_2^* \neq \emptyset$. Otherwise, if one of the slabs contains no point of S, we can divide the other slab into two halves, obtaining a pair of parallel slabs of width $w^*/2$ whose union still covers S, contradicting the optimality.

Let $Q \subseteq S$ be an ε-certificate of S (see Section 3.3). Then we have the following.

Lemma 37. *$Q \cap \sigma_1^* \neq \emptyset$ and $Q \cap \sigma_2^* \neq \emptyset$.*

Proof. Suppose $Q \cap \sigma_1^* = \emptyset$. That is, $Q \subseteq S \cap \sigma_2^*$. Let $\Sigma = (\tau_1, \tau_2)$ be the pair of slabs obtained by dividing σ_2^* into two halves. See Figure 25(a). That is $w(\tau_1) = w(\tau_2) = w(\sigma_2^*)/2$. Observe that $Q \subset \tau_1 \cup \tau_2$. Let $\Sigma' = (\tau_1', \tau_2')$ be the pair of slabs where τ_1' and τ_2' are the ε-expansions of τ_1 and τ_2, respectively. Let D be the amount by which each bounding line is shifted by the ε-expansion. That is, $D = (1+\varepsilon)w(\tau_1)/2 = (1+\varepsilon)w(\tau_2)/2 \leq (1+\varepsilon)w^*/4$. See Figure 25(b).

Let p a point in $S \cap \sigma_1^*$. Without loss of generality, assume that τ_1 is closer to p than τ_2. We now show that τ_1' does not cover p (which also implies that τ_2' does not cover p), contradicting that Q is an ε-certificate of S. We have $d(p, \tau_1) \geq g(\Sigma^*) \geq \rho W(\Sigma^*) \geq \rho w^*$. Since $\rho \geq 1/2 > (1+\varepsilon)/4$, it follows that $d(p, \tau_1) \geq \rho w^* > (1+\varepsilon)w^*/4$. As shown above, $D \leq (1+\varepsilon)w^*/4$. Therefore, τ_1 does not cover p.

Symmetrically, we can show that $Q \cap \sigma_2^* = \emptyset$ leads to a contradiction. □

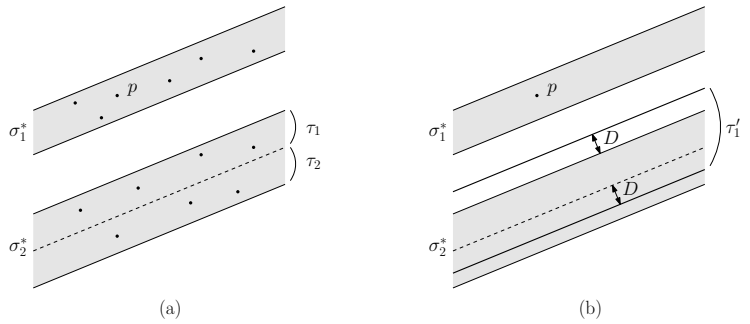


Figure 25: (a) Black points indicate the point in S. The dashed line indicates the center line of σ_2^*. (b) The ε-expansion τ_1' of τ_1 is shown.

Now we prove Theorem 30. By Lemma 37, $\Sigma^* = (\sigma_1^*, \sigma_2^*)$ satisfies the three conditions in the statement of Theorem 36 with respect to Q. Therefore, we obtain a pair of parallel slabs by running the algorithm in Theorem 36 on Q. Let $\Sigma = (\sigma_1, \sigma_2)$ be the pair. (If the assumption on the gap ratio is not satisfied, the algorithm in Theorem 36 may return NO. In that case, we stop the algorithm and also return NO.) Observe that $\max\{w(\sigma_1), w(\sigma_2)\} \leq w^*$ since $Q \subseteq S$. Let $\Sigma' = (\sigma_1', \sigma_2')$ be the pair of slabs where σ_1' and σ_2' are the ε-expansions of σ_1 and σ_2, respectively. By definition, $\sigma_1' \cup \sigma_2'$ covers S and $\max\{w(\sigma_1'), w(\sigma_2')\} \leq (1+\varepsilon)w^*$. Therefore, $\Sigma' = (\sigma_1', \sigma_2')$ forms a $(1+\varepsilon)$-approximate solution.

Computing $Q \subseteq S$ of size $O(1/\varepsilon^2)$ can be done in $O(n)$ time by Theorem 8. It takes $O(|Q| \log |Q|)$ time to obtain (σ_1', σ_2') by Theorem 36. Let $\mu = \min\{n, 1/\varepsilon^2\}$, then we have $|Q| = O(\mu)$. Therefore, the total running time is $O(n + \mu \log \mu)$, which can be expressed as both $O(n \log(1/\varepsilon))$ and $O(n + (1/\varepsilon^2)\log(1/\varepsilon))$.

8 Future Work

We have presented approximation algorithms for several variants of the two-line-center problem. An open question is whether the dependence on ε can be further refined for each variant. For the exact algorithms of the problems, no nontrivial lower bounds are currently known, except for the one-fixed-orientation case, for which we provide an optimal algorithm.

Each of the four problems admits natural generalizations with respect to the dimension d and the number of lines k (our problems correspond to $d = 2$ and $k = 2$). For higher dimensions ($d \geq 3$) or a larger number of lines ($k \geq 3$), very few exact or approximation algorithms are known. For the general k-line-center problem, the authors in [3] gave a $(1+\varepsilon)$-approximation algorithm whose expected running time is $O(n \log n)$ where the constant depends on d, k, and ε. We leave open the question of whether efficient exact and approximation algorithms can be obtained for these and other variants of the two-line-center problem in higher dimensions or with more than two lines.

Another natural variant of the two-line-center problem is the following. Given a set S of n points in the plane and a fixed orientation difference θ, the goal is to find a pair of slabs whose orientations differ by θ, whose union covers S, and that minimizes the maximum of their widths. An exact algorithm with running time $O(n^2 \alpha(n) \log n)$ is known for this problem [6], where $\alpha(n)$ denotes the inverse Ackermann function. To the best of our knowledge, no approximation algorithm is known for this variant. It is unclear if our approach based on an anchor pair of an optimal pair of slabs and a constant factor solution (as in Lemma 20 and Lemma 34) can be extended to this variant. This is because rotating one slab requires rotating the other as well. We addressed this issue in the parallel two-line-center problem by considering two different cases, depending on the gap-ratio of an optimal pair of slabs (see Sections 7.1 and 7.2).

References

[1] Pankaj K. Agarwal and Cecilia Magdalena Procopiuc. Approximation algorithms for projective clustering. *J. Algorithms*, 46(2):115–139, 2003. doi:10.1016/S0196-6774(02)00295-X.

[2] Pankaj K. Agarwal, Cecilia Magdalena Procopiuc, and Kasturi R. Varadarajan. A $(1+\varepsilon)$-approximation algorithm for 2-line-center. *Comput. Geom.*, 26(2):119–128, 2003. doi:10.1016/S0925-7721(03)00017-8.

[3] Pankaj K. Agarwal, Cecilia Magdalena Procopiuc, and Kasturi R. Varadarajan. Approximation algorithms for a k-line center. *Algorithmica*, 42(3-4):221–230, 2005. doi:10.1007/S00453-005-1166-X.

[4] Charu C. Aggarwal, Cecilia Magdalena Procopiuc, Joel L. Wolf, Philip S. Yu, and Jong Soo Park. Fast algorithms for projected clustering. In *SIGMOD 1999, Proceedings ACM SIGMOD International Conference on Management of Data, June 1-3, 1999, Philadelphia, Pennsylvania, USA*, pages 61–72. ACM Press, 1999. doi:10.1145/304182.304188.

[5] Charu C. Aggarwal and Philip S. Yu. Finding generalized projected clusters in high dimensional spaces. In *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data, May 16-18, 2000, Dallas, Texas, USA*, pages 70–81. ACM, 2000. doi:10.1145/342009.335383.

[6] Taehoon Ahn and Sang Won Bae. Constrained two-line center problems. In *35th International Symposium on Algorithms and Computation, ISAAC 2024, Sydney, Australia, December 8-11, 2024*, volume 322 of *LIPIcs*, pages 5:1–5:17. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2024. doi:10.4230/LIPICS.ISAAC.2024.5.

[7] Mihai Badoiu, Sariel Har-Peled, and Piotr Indyk. Approximate clustering via core-sets. In *Proceedings on 34th Annual ACM Symposium on Theory of Computing, May 19-21, 2002, Montréal, Québec, Canada*, pages 250–257. ACM, 2002. doi:10.1145/509907.509947.

[8] Sang Won Bae. Minimum-width double-strip and parallelogram annulus. *Theor. Comput. Sci.*, 833:133–146, 2020. doi:10.1016/J.TCS.2020.05.045.

[9] Gill Barequet and Sariel Har-Peled. Efficiently approximating the minimum-volume bounding box of a point set in three dimensions. *J. Algorithms*, 38(1):91–109, 2001. doi:10.1006/JAGM.2000.1127.

[10] Manuel Blum, Robert W. Floyd, Vaughan R. Pratt, Ronald L. Rivest, and Robert Endre Tarjan. Time bounds for selection. *J. Comput. Syst. Sci.*, 7(4):448–461, 1973. doi:10.1016/S0022-0000(73)80033-9.

[11] Kaushik Chakrabarti and Sharad Mehrotra. Local dimensionality reduction: A new approach to indexing high dimensional spaces. In *VLDB 2000, Proceedings of 26th International Conference on Very Large Data Bases, September 10-14, 2000, Cairo, Egypt*, pages 89–100. Morgan Kaufmann, 2000. URL: http://www.vldb.org/conf/2000/P089.pdf.

[12] Timothy M. Chan. Faster core-set constructions and data-stream algorithms in fixed dimensions. *Comput. Geom.*, 35(1-2):20–35, 2006. doi:10.1016/J.COMGEO.2005.10.002.

[13] Chaeyoon Chung, Taehoon Ahn, Sang Won Bae, and Hee-Kap Ahn. Parallel line centers with guaranteed separation. *Comput. Geom.*, 129:102185, 2025. doi:10.1016/J.COMGEO.2025.102185.

[14] Vasek Chvátal. A greedy heuristic for the set-covering problem. *Math. Oper. Res.*, 4(3):233–235, 1979. doi:10.1287/MOOR.4.3.233.

[15] Martin E. Dyer. Linear time algorithms for two- and three-variable linear programs. *SIAM J. Comput.*, 13(1):31–45, 1984. doi:10.1137/0213003.

[16] Greg N. Frederickson and Donald B. Johnson. The complexity of selection and ranking in X+Y and matrices with sorted columns. *J. Comput. Syst. Sci.*, 24(2):197–208, 1982. doi:10.1016/0022-0000(82)90048-4.

[17] Greg N. Frederickson and Donald B. Johnson. Generalized selection and ranking: Sorted matrices. *SIAM J. Comput.*, 13(1):14–30, 1984. doi:10.1137/0213002.

[18] Sariel Har-Peled and Kasturi R. Varadarajan. Projective clustering in high dimensions using core-sets. In *Proceedings of the 18th Annual Symposium on Computational Geometry, Barcelona, Spain, June 5-7, 2002*, pages 312–318. ACM, 2002. doi:10.1145/513400.513440.

[19] Refael Hassin and Nimrod Megiddo. Approximation algorithms for hitting objects with straight lines. *Discret. Appl. Math.*, 30(1):29–42, 1991. doi:10.1016/0166-218X(91)90011-K.

[20] Michael E. Houle and Godfried T. Toussaint. Computing the width of a set. *IEEE Trans. Pattern Anal. Mach. Intell.*, 10(5):761–765, 1988. doi:10.1109/34.6790.

[21] Piotr Indyk, Rajeev Motwani, Prabhakar Raghavan, and Santosh S. Vempala. Locality-preserving hashing in multidimensional spaces. In *Proceedings of the Twenty-Ninth Annual ACM Symposium on the Theory of Computing, El Paso, Texas, USA, May 4-6, 1997*, pages 618–625. ACM, 1997. doi:10.1145/258533.258656.

[22] Jerzy W. Jaromczyk and Miroslaw Kowaluk. The two-line center problem from a polar view: A new algorithm and data structure. In *Algorithms and Data Structures, 4th International Workshop, WADS '95, Kingston, Ontario, Canada, August 16-18, 1995, Proceedings*, volume 955 of *Lecture Notes in Computer Science*, pages 13–25. Springer, 1995. doi:10.1007/3-540-60220-8_47.

[23] D. T. Lee and Ying-Fung Wu. Geometric complexity of some loction problems. *Algorithmica*, 1(2):193–211, 1986. doi:10.1007/BF01840442.

[24] Nimrod Megiddo. Linear programming in linear time when the dimension is fixed. *J. ACM*, 31(1):114–127, 1984. doi:10.1145/2422.322418.

[25] Nimrod Megiddo and Arie Tamir. On the complexity of locating linear facilities in the plane. *Oper. Res. Lett.*, 1(5):194–197, 1982. doi:10.1016/0167-6377(82)90039-6.

[26] Franco P. Preparata and Michael Ian Shamos. *Computational Geometry - An Introduction*. Texts and Monographs in Computer Science. Springer, New Haven, CT, USA, 1985. UMI Order No. 78-18546. doi:10.1007/978-1-4612-1098-6.

[27] Cecilia M. Procopiuc. Projective clustering. In *Encyclopedia of Machine Learning and Data Mining*, pages 1018–1025. Springer, 2017. doi:10.1007/978-1-4899-7687-1_676.

[28] Cecilia Magdalena Procopiuc, Michael Jones, Pankaj K. Agarwal, and T. M. Murali. A monte carlo algorithm for fast projective clustering. In *Proceedings of the 2002 ACM SIGMOD International Conference on Management of Data, Madison, Wisconsin, USA, June 3-6, 2002*, pages 418–427. ACM, 2002. doi:10.1145/564691.564739.

[29] Haitao Wang. Dynamic convex hulls under window-sliding updates. *Comput. Geom. Topol.*, 4(1):3:1–3:14, 2025. doi:10.57717/CGT.V4I1.53.

A Proof of Theorem 5

In this section, we give a 10-approximation algorithm for the general two-line-center problem that runs in $O(n)$ time. As mentioned in the main text, we use the following $O(n)$-time algorithm that incrementally computes a 6-approximation[5] of the width of a point set.

Theorem 4 (Theorem A.3 of [12])**.** *Given an online sequence of points in the plane, we can maintain a* 6*-approximation of the width with* $O(1)$ *space and update time.*

The algorithm from Theorem 4 maintains the following throughout the insertion of points: a set V of points, a real value w, and the first inserted point o. Without loss of generality, assume that o is the origin. It initializes $V = \emptyset$ and $w = 0$. Whenever a new point p is inserted, V and w are updated according to the procedure described in Algorithm 1.

In [12], it is shown that the size of V remains $O(1)$ at any time, and that the final value of w, denoted by w_f, after inserting a set P of points, satisfies $\text{width}(P) \leq w_f \leq 6 \cdot \text{width}(P)$.

Algorithm 1 Processing a new point p (See the proof of Theorem A.3 in [12]. Let $\delta > 0$ be a sufficiently small constant.)

```
1: function INSERT(p)
2:     w = max{w, 6 · width(V ∪ {o, p})}
3:     if for all v ∈ V , ‖p‖ > (1 + δ)‖v‖ or ∠pov > π/2 + δ then
4:         Insert p to V
5:         Remove all points v from V such that ‖v‖ ≤ δ‖p‖ and ∠pov ≤ π/2+δ
6:     end if
7: end function
```

By Lemma 2, we can find a set F of at most 11 pairs of points, at least one of which is an anchor pair of Σ^* in $O(n)$ time. For the moment, assume that we have an anchor pair (p, q) of Σ^*. (We will repeat the following algorithm for every pair in F.) Without loss of generality, assume that $p, q \in \sigma_1^*$ and $d(p, q) \geq \text{diam}((\sigma_1^* \setminus \sigma_2^*) \cap S)/4$.

Our Algorithm For $1 \leq i \leq n$, let $f(i) = w(\sigma(p, q; p_i)) = 2d(\ell_{pq}, p_i)$ and let $g(i)$ denote a 6-approximate width of $S \setminus \sigma(p, q; p_i)$ obtained by Theorem 4. The value $g(i)$ depends on the order in which points are inserted, but this does not affect the correctness of our algorithm. Observe that $f(i)$ is an increasing function, whereas $g(i)$ is not necessarily decreasing (see Figure 26(a)). Nevertheless, we proceed as in Naive Approach and Idea.

In the following, we describe our algorithm, which is also summarized in Algorithm 2. We then prove its correctness in Lemma 42 and analyze its running time in Theorem 5.

Data Structures and Invariants. Throughout the algorithm, we maintain the following: a range of indices $R \subseteq [1, n]$, a set $Q \subseteq S$, a value w, and

[5]The algorithm in [12] maintains a $(5+\varepsilon)$-approximation of the minimum width for a fixed $\varepsilon > 0$. For simplicity, we maintain a 6-approximation.

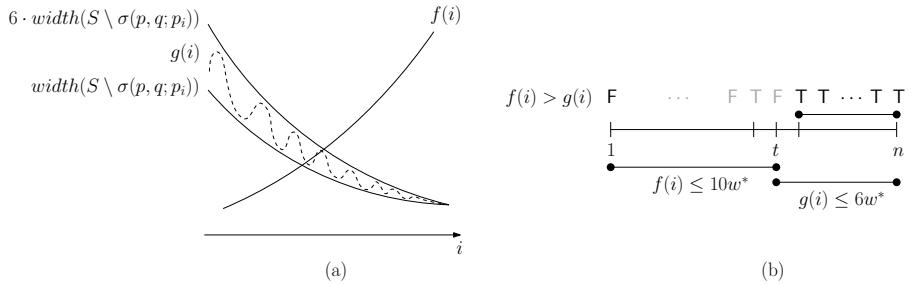


Figure 26: (a) A rough illustration of $f(i)$ and $g(i)$. $f(i)$ is an increasing function, whereas $g(i)$ is not necessarily monotone. (b) An illustration of Lemma 39.

a set V. Initially, we set $R = [1, n]$, $Q = S$, $w = 0$, and $V = \emptyset$. We also set $o = p_n$, which is used in Algorithm 1. Without loss of generality, assume that o is the origin.

We maintain these variables so that the following invariants hold: (1) $Q = \{p_i \mid i \in R\}$, and (2) w and V are the value and the set obtained by inserting the points[6] of $S \setminus \sigma(p, q; p_e)$ using Algorithm 1, where e is the largest index in R. That is, $\mathrm{width}(S \setminus \sigma(p, q; p_e)) \leq w \leq 6 \cdot \mathrm{width}(S \setminus \sigma(p, q; p_e))$.

Main loop. We perform a binary search on the index range R. At each step, let $R = [s, e]$, set m to be the median of R, and find the point p_m. We then compare $f(m)$ and $g(m)$. The value $f(m)$ can be computed directly from the coordinates of p, q, and p_m. To obtain $g(m)$, we need to insert the points of $S \setminus \sigma(p, q; p_m)$ using Algorithm 1. However, since we already have w and V obtained by inserting the points in $S \setminus \sigma(p, q; p_e)$, and $S \setminus \sigma(p, q; p_e) \subseteq S \setminus \sigma(p, q; p_m)$, we can obtain $g(m)$ by inserting only the additional points $(S \setminus \sigma(p, q; p_m)) \setminus (S \setminus \sigma(p, q; p_e)) = \{p_i | m < i \leq e\}$.

If $f(m) \geq g(m)$, we set $R = [s, m]$. In this case, since the largest index in R changes, we update w and V to maintain the invariant, using the value and the set obtained when computing $g(m)$ (see line 11 of Algorithm 2). If $f(m) < g(m)$, we set $R = [m, e]$ and w and V remain unchanged. In both cases, we update Q to reflect the new range R.

Base case. When R contains at most three indices, we select an index $r \in R$ that minimizes $\max\{f(r), g(r)\}$. Observe that for every $i \in R$, $f(i)$ is computed directly and $g(i)$ can be computed by inserting at most three points using Algorithm 1. Finally, we output the pair of slabs, where one slab is $\sigma(p, q; p_r)$ and the other is a slab of width at most $g(r)$ that covers $S \setminus \sigma(p, q; p_r)$. We note that the latter slab can be obtained by o, w, and V (see the proof of Theorem A.3 in [12]).

To prove the correctness in Lemma 42, we first show Lemma 38, 39, and Corollary 40, 41.

▶ **Lemma 38.** *For $R = [s, e]$ at any time, it holds that $f(s) < g(s)$ and $f(e) \geq g(e)$.*

Proof. Let $I(i)$ be the boolean value indicating whether $f(i) \geq g(i)$. We will prove that $I(s) = \mathsf{False}$ and $I(e) = \mathsf{True}$ for $R = [s, e]$ throughout the algorithm.

[6]Observe that p_n is always contained in $S \setminus \sigma(p, q; p_e)$ except for when $e = n$. This is why we set $o = p_n$.

Algorithm 2 A linear-time 10-approximation algorithm

1: $R \leftarrow [1, n]$
2: $Q \leftarrow S$
3: $w \leftarrow 0,\ V \leftarrow \emptyset,\ o \leftarrow p_n$
4: **while** $|R| > 3$ (equivalently, $|Q| > 3$) **do**
5: $\quad m \leftarrow \lfloor (s + e)/2 \rfloor$, where $R = [s, e]$
6: $\quad$ Find the point p_m in Q
7: $\quad f \leftarrow 2d(\ell_{pq}, p_m)$
8: $\quad g, V' \leftarrow$ a 6-approximate width of $S \setminus \sigma(p, q; p_m)$ and a set obtained by Theorem 4
9: $\quad$ **if** $f \geq g$ **then**
10: $\quad\quad R \leftarrow [s, m]$
11: $\quad\quad (w, V) \leftarrow (g, V')$
12: $\quad\quad Q \leftarrow \{p_s, p_{s+1}, \ldots, p_m\}$
13: $\quad$ **else**
14: $\quad\quad R \leftarrow [m, e]$
15: $\quad\quad Q \leftarrow \{p_m, p_{m+1}, \ldots, p_e\}$
16: $\quad$ **end if**
17: **end while**
18: Find an index $r \in R$ which minimizes $\max\{f(r), g(r)\}$
19: Report the pair of slabs corresponding to p_r: one is $\sigma(p, q; p_r)$, and the other is a slab of width at most $g(r)$ that covers $S \setminus \sigma(p, q; p_r)$.

We prove this by induction on the iteration of the while loop. Initially, $R = [1, n]$. Observe that $f(1) = 2d(\ell_{pq}, p_1) = 0$. Since $g(1) > 0$, we have $f(1) < g(1)$, so $I(1) = \mathsf{False}$. On the other hand, $g(n) = 0$ because $S \setminus \sigma(p, q; p_n) = \emptyset$, which implies $I(n) = \mathsf{True}$.

Now, let $R = [s_i, e_i]$ be the range of indices at the beginning of the i-th iteration of the while loop, and let $m_i = \lfloor (s_i + e_i)/2 \rfloor$. Assume as the induction hypothesis that $I(s_i) = \mathsf{False}$ and $I(e_i) = \mathsf{True}$. Let s_{i+1} and e_{i+1} be the indices defining the range after this iteration.

If $I(m_i) = \mathsf{True}$, then $[s_{i+1}, e_{i+1}] = [s_i, m_i]$ (see line 10). Otherwise, $[s_{i+1}, e_{i+1}] = [m_i, e_i]$ (see line 14). In both cases, it holds that $I(s_{i+1}) = \mathsf{False}$ and $I(e_{i+1}) = \mathsf{True}$. □

Let t be the largest index such that $f(t) = 2d(\ell_{pq}, p_t) \leq 10w^*$. As $d(\ell_{pq}, p_1) = 0$, t exists.

▶ **Lemma 39.** *The following statements hold. See Figure 26(b).*

(i) For every i with $1 \leq i \leq t$, we have $f(i) \leq 10w^*$.

(ii) For every i with $t \leq i \leq n$, we have $g(i) \leq 6w^*$.

(iii) For every i with $t < i \leq n$, we have $f(i) > g(i)$.

Proof. **(i)** This follows directly from the definition of t and the fact that $f(i)$ is an increasing function.

(ii) Let k be the smallest index such that $\sigma(p, q; p_k)$ covers the points in $S \cap (\sigma_1^* \setminus \sigma_2^*)$. By Lemma 3, we have $2d(\ell_{pq}, p_k) \leq 10w^*$. Therefore, $k \leq t$ by the definition of t. Observe that $k \leq i$ for every i with $t \leq i \leq n$.

By the definition of $g(i)$, we have $g(i) \leq 6 \cdot \text{width}(S \setminus \sigma(p, q; p_i))$. Since $k \leq i$, we also have $\text{width}(S \setminus \sigma(p, q; p_i)) \leq \text{width}(S \setminus \sigma(p, q; p_k))$. Therefore, $g(i) \leq 6 \cdot \text{width}(S \setminus \sigma(p, q; p_k))$. Finally, $\text{width}(S \setminus \sigma(p, q; p_k)) \leq w^*$, otherwise $w(\sigma_2^*) \leq w^*$ would be contradicted because $\sigma(p, q; p_k)$ covers all points in $S \cap (\sigma_1^* \setminus \sigma_2^*)$. Therefore, $g(i) \leq 6w^*$ for every i with $t \leq i \leq n$.

(iii) By the definition of t, we have $10w^* < 2d(\ell_{pq}, p_i) = f(i)$ for every i with $t < i \leq n$. From (ii), we know that $g(i) \leq 6w^*$ in the same range. Thus, for all i with $t < i \leq n$, we have $g(i) \leq 6w^* \leq 10w^* < f(i)$. □

By combining Lemma 39(i) and (ii), we obtain the following corollary.

Corollary 40. $\max\{f(t), g(t)\} \leq 10w^*$.

Let R_f be the final range R obtained after the while loop.

▶ **Corollary 41.** *There exists $j \in R_f$ such that $j \leq t$.*

Proof. Suppose, to the contrary, that all indices in R_f are at least $t + 1$. By Lemma 39(iii), we have $f(i) > g(i)$ for every $i \in R_f$. In particular, if $R_f = [s, e]$, then $f(s) > g(s)$, which contradicts Lemma 38. □

▶ **Lemma 42.** *For the final index $r \in R_f$ obtained in base case,* $\max\{f(r), g(r)\} \leq 10w^*$.

Proof. If $t \in R_f$, then $\max\{f(r), g(r)\} \leq \max\{f(t), g(t)\} \leq 10w^*$ by Corollary 40.

If $t \notin R_f$, then Corollary 41 implies that $R_f \subseteq [1, t - 1]$. By Lemma 38, there exists an index $b \in R_f$ such that $f(b) \geq g(b)$. Hence, $\max\{f(b), g(b)\} = f(b) \leq f(t) \leq \max\{f(t), g(t)\} \leq 10w^*$. Therefore, we have $\max\{f(r), g(r)\} \leq \max\{f(b), g(b)\} \leq 10w^*$. □

Now we show the following theorem.

Theorem 5. *In $O(n)$ time, we can compute a pair of slabs that forms a 10-approximation for the general two-line-center problem.*

Proof. By Lemma 42, our algorithm computes a pair of slabs whose widths are at most $10w^*$ and whose union covers S. We now analyze its running time.

Initializing the variables R, Q, w, V, and o clearly can be done in $O(n)$ time.

Next, we show that each iteration of the while loop, with the current index range $R = [s, e]$, takes $O(|R|)$ time. For the median m of R, we find p_m using a linear-time selection algorithm in Q, which can be done in $O(|Q|) = O(|R|)$ time [10].

The value $f(m)$ is obtained directly. To compute $g(m)$, we apply Algorithm 1 to the points $(S \setminus \sigma(p, q; p_m)) \setminus (S \setminus \sigma(p, q; p_e)) = \{p_i | m < i \leq e\}$ updating the current w and V. By Theorem 4, this takes $O(e - m) = O(|R|)$ time.

Since the size of R decreases by a factor of two in each iteration, the total time over all iterations of the while loop is $O(n + n/2 + n/4 + \cdots) = O(n)$.

After the while loop ends, we find an index $r \in R$ which minimizes $\max\{f(r), g(r)\}$. Observe that for every $i \in R$, $f(i)$ is obtained directly and $g(i)$ can be computed by inserting at most three points using Algorithm 1. Therefore, it takes $O(1)$ time for finding r. We obtain one slab $\sigma(p, q; p_r)$ and it remains to compute the other slab, which has width at most $g(r)$ and covers

$S \setminus \sigma(p, q; p_r)$. In the proof of Theorem A.3 in [12], it is shown that this can be done by computing the minimum-width slab enclosing $V \cup \{o\}$ and expanding it by a factor determined by δ and w. Since $|V| = O(1)$ at all times, this step takes $O(|V| \log |V|) = O(1)$ time.

We resolve the assumption that we have an anchor pair (p, q) of Σ^*. We repeat the entire algorithm for every pair in F, using it as (p, q), which adds only a constant factor to the running time. Observe that for each pair in F, we obtain a pair of slabs whose union covers S. Among these pairs, we output the one with the smallest maximum width. □

B Proof of Theorem 18

We first sort the points of S so that $s_1, \ldots, s_n$ are ordered in decreasing order of their y-coordinates. Let W denote the set of all pairwise differences between the y-coordinates of points in S. Since W can be represented as a sorted matrix [16], we can identify two consecutive values $w_0, w_1 \in W$ such that $w_0 < w^* \leq w_1$ in $O(n \log n)$ time, by combining our decision algorithm described in Section 4.2 with an efficient selection algorithm for sorted matrices [17].

Notation We use the same notation S_a, S_b, and $\{\ell_i, \theta_i, a_i, b_i \mid i = 1, 2\}$ as in Section 4.2. For any $w > 0$, we denote by σ_w a horizontal slab of width w.

Assumption Throughout this section, we assume that there exists a position of σ_{w^*} such that $\mathrm{width}(S \setminus \sigma_{w^*}) \leq w^*$ (i.e., $\mathrm{width}(S_a \cup S_b) \leq w^*$) and S_a dominates S_b in that position. The other case can be handled symmetrically by reflecting S across a horizontal line.

We first introduce two lemmas from [6]. Observe that the sequence of changes in the sets S_a and S_b that occur as σ_w moves downward is the same for any $w_0 < w < w_1$. Let X denote the set of all pairs (A, B) such that $A \subseteq S$ and $B \subseteq S$ appear as S_a and S_b, respectively, during the sweep of the plane by σ_w for $w_0 < w < w_1$.

Lemma 43 (Lemma 9 of [6])**.** $w^* = \min\{w_1, \min_{(S_a, S_b) \in X}\{\mathrm{width}(S_a \cup S_b)\}\}$.

Lemma 43 implies that by tracking $\mathrm{width}(S_a \cup S_b)$ as σ_w moves downward (for any $w_0 < w < w_1$), we can determine w^* by comparing its minimum value with w_1. Lemma 44 and Corollary 45 further refine this observation, implying that w^* can be obtained more directly from the procedure of the decision algorithm.

Lemma 44 (Lemma 11 of [6])**.** *For $(S_a, S_b) \in X$ where S_a dominates S_b, it holds that* $\mathrm{width}(S_a \cup S_b) < w_1$ *if and only if* $\mathrm{width}_{[\theta_1, \theta_2]}(S_a) < w_1$*. Moreover, if* $\mathrm{width}_{[\theta_1, \theta_2]}(S_a) < w_1$*, then* $\mathrm{width}(S_a \cup S_b) = \mathrm{width}_{[\theta_1, \theta_2]}(S_a)$.

By combining Lemma 43 and 44, we can obtain the following corollary.

Corollary 45. $w^* = \min\{w_1, \min\{\mathrm{width}_{[\theta_1, \theta_2]}(S_a) \mid (S_a, S_b) \in X,\ S_a \text{ dominates } S_b\}\}$.

By Corollary 45, once we find a position of σ_w where S_a dominates S_b and $\mathrm{width}_{[\theta_1, \theta_2]}(S_a)$ is minimized, we can compute w^* by comparing this value with w_1. We now show that we can find the position during the procedures in the decision algorithm presented in Section 4.2.

We first fix $w_0 < w < w_1$. Recall that in the decision algorithm, we test whether $\text{width}_{[\theta_1,\theta_2]}(S_a) < w$ while σ_w moves downward, maintaining the following data structures (see *Data structures and Main loop* in Section 4.2).

- the dynamic data structures for $\text{conv}(S_a)$ and $\text{conv}(S_b)$ (from Lemma 12)
- the tangent lines ℓ_1 and ℓ_2, with their corresponding points a_1, b_1, a_2, and b_2
- the two antipodal pairs of S_a with respect to the orientations θ_1 and θ_2, respectively

As σ_w moves downward, we will maintain an additional value $w_{\min}$, which represents the minimum value of $\text{width}_{[\theta_1,\theta_2]}(S_a)$ observed so far. Initially, we set $w_{\min} = \infty$. Observe that S_b dominates S_a from the beginning until a certain moment, after which S_a dominates S_b until the end. Until S_a begins to dominate S_b (which can be easily detected using Observation 11), we just follow the decision algorithm without updating $w_{\min}$. At the moment when S_a starts to dominate S_b, we set $w_{\min} = \text{width}_{[\theta_1,\theta_2]}(S_a)$. This can be done in $O(n)$ time by following the antipodal pairs in the data structure that we maintain for $\text{conv}(S_a)$.

In the following, we show how $w_{\min}$ can be updated at each event during the execution of the decision algorithm. At the end of the procedure, the minimum between w_1 and the final value of $w_{\min}$ is equal to w^* by Corollary 45.

We note that the authors of [6] also used the observation from Corollary 45 to perform a sweep of the slab σ_w. However, their approach requires an additional data structure to compute $\text{width}_{[\theta_1,\theta_2]}(S_a)$ at each event, resulting in $O(\log^3 n)$ time per event.

Insertion into S_a Let p be the point inserted into S_a, and let $S'_a = S_a \cup \{p\}$. We define $\{\ell'_i, \theta'_i, a'_i, b'_i \mid i = 1, 2\}$ analogously for S'_a and S_b. That is ℓ'_1 and ℓ'_2 are the right and the left outer common tangents of S'_a and S_b, respectively. Let $w'_{\min}$ denote the updated value after inserting p. Therefore, $w'_{\min} = \min\{w_{\min}, \text{width}_{[\theta'_1,\theta'_2]}(S'_a)\}$.

Suppose that p lies on or between ℓ_1 and ℓ_2. Then both ℓ_1 and ℓ_2 remain unchanged, and consequently so do θ_1 and θ_2. Since every slab $\sigma_\theta(S_a)$ for $\theta \in [\theta_1, \theta_2]$ contains p, it follows that $\text{width}_{[\theta_1,\theta_2]}(S_a) = \text{width}_{[\theta_1,\theta_2]}(S'_a)$. As $w_{\min} \leq \text{width}_{[\theta_1,\theta_2]}(S_a)$, $w'_{\min} = w_{\min}$.

Now assume that p lies to the right of ℓ_1 (the case where p lies to the left of ℓ_2 can be handled symmetrically). In this case, $\ell'_2 = \ell_2$, while $\ell'_1 \neq \ell_1$. Observe that $\theta'_2 = \theta_2$ and $\theta'_1 < \theta_1$. Now we show the following.

▷ Claim 46. $w'_{\min} = \min\{w_{\min}, \text{width}_{[\theta'_1,\theta_1]}(S'_a)\}$

Proof. Let $a = \text{width}_{[\theta'_1,\theta_1]}(S'_a)$, $b = \text{width}_{[\theta_1,\theta_2]}(S'_a)$, and $c = \text{width}_{[\theta'_1,\theta_2]}(S'_a)$. By definition, $c = \min\{a, b\}$. As $w'_{\min} = \min\{w_{\min}, c\}$, we have $w'_{\min} = \min\{w_{\min}, a, b\}$. Observe that $w_{\min} \leq \text{width}_{[\theta_1,\theta_2]}(S_a) \leq b$ by the definition of $w_{\min}$ and $S_a \subset S'_a$. Therefore, we have $w'_{\min} = \min\{w_{\min}, a\} = \min\{w_{\min}, \text{width}_{[\theta'_1,\theta_1]}(S'_a)\}$. ◁

The above claim allows us to consider only the range $[\theta'_1, \theta_1]$. Recall that, in *Testing the width condition* step of the decision algorithm for insertions,

the antipodal pairs of S'_a corresponding to the orientation range $[\theta'_1, \theta_1]$ are visited in order. Therefore, $\text{width}_{[\theta'_1,\theta_1]}(S'_a)$ can be obtained directly during this procedure.

Deletion from S_b Let p be the point deleted from S_b, and let $S'_b = S_b \setminus \{p\}$. We define $\{\ell'_i, \theta'_i, a'_i, b'_i \mid i = 1, 2\}$ analogously for S_a and S'_b. That is ℓ'_1 and ℓ'_2 are the right and the left outer common tangents of S_a and S'_b, respectively. Let $w'_{\min}$ denote the updated value after deleting p. Therefore, $w'_{\min} = \min\{w_{\min}, \text{width}_{[\theta'_1,\theta'_2]}(S_a)\}$.

Suppose that p lies strictly between ℓ_1 and ℓ_2. Then both ℓ_1 and ℓ_2 remain unchanged, and consequently so do θ_1 and θ_2. Therefore, $w'_{\min} = w_{\min}$ by definition.

Now assume that p lies on or to the right of ℓ_1 (the case where p lies on or to the left of ℓ_2 can be handled symmetrically). In this case, $\ell'_2 = \ell_2$, while $\ell'_1 \neq \ell_1$. Observe that $\theta'_2 = \theta_2$ and $\theta'_1 < \theta_1$. Therefore, $w'_{\min} = \min\{w_{\min}, \text{width}_{[\theta'_1,\theta_2]}(S_a)\} = \min\{w_{\min}, \text{width}_{[\theta'_1,\theta_1]}(S_a), \text{width}_{[\theta_1,\theta_2]}(S_a)\}$. Note that $w_{\min} \leq \text{width}_{[\theta_1,\theta_2]}(S_a)$. Therefore, $w'_{\min} = \min\{w_{\min}, \text{width}_{[\theta'_1,\theta_1]}(S_a)\}$.

The above observation allows us to consider only the range $[\theta'_1, \theta_1]$. Recall that, in *Testing the width condition* step of the decision algorithm for deletions, the antipodal pairs of S_a corresponding to the orientation range $[\theta'_1, \theta_1]$ are visited in order. Therefore, $\text{width}_{[\theta'_1,\theta_1]}(S_a)$ can be obtained directly during this procedure.

Now we can conclude the following theorem.

Theorem 18. *Given a set S of n points in the plane and an orientation θ, we can find a pair of slabs such that one of them has orientation θ and their union covers S while minimizing the maximum of their widths in $O(n \log n)$ time.*

C Lower Bound for the One-Fixed-Orientation Problem

▶ **Lemma 47.** *The one-fixed-orientation two-line-center problem requires $\Omega(n \log n)$ time in the algebraic computation tree model.*

Proof. Given a set P of n points in the plane, it is known that computing $\text{width}(P)$ requires $\Omega(n \log n)$ time [20, 23]. We give a reduction from the width problem to the one-fixed-orientation two-line-center problem. We show that we can construct a set Q of $n+2$ points in $O(n)$ time such that the following holds: Let (σ^*_h, σ^*_t) be a pair of slabs whose union covers Q, where σ^*_h is horizontal, and the maximum of their widths is minimized. Let $w^*(Q) = \max\{w(\sigma^*_h), w(\sigma^*_t)\}$. Then $w^*(Q) = \text{width}(P)$.

We compute the largest difference between the x-coordinates of the points in P, and denote it by L_1. Similarly, we compute the largest difference between the y-coordinates of the points in P, and denote it by L_2. Let $L = \max\{L_1, L_2\}$.

Consider a square of side length L that covers P, and denote it by R. Let $q_1, q_2 \in \mathbb{R}^2$ be two points on a horizontal line such that $d(q_1, q_2) = 7L$, $d(R, \ell_{q_1 q_2}) = L$, and R and $\overline{q_1 q_2}$ are vertically center-aligned (see Figure 27). Let $Q = P \cup \{q_1, q_2\}$.

It is clear that $w^*(Q) \leq \text{width}(P)$, as we can construct a pair of slabs of width width(P) where one covers P and the other covers $\{q_1, q_2\}$. Thus, it remains to show that $w^*(Q) \geq \text{width}(P)$. Assume to the contrary that $w^*(Q) < \text{width}(P)$. Since q_1 and q_2 lie on a horizontal line, we have two cases: (1) $q_1, q_2 \in \sigma_h^*$, or (2) $q_1, q_2 \notin \sigma_h^*$.

Consider the case where $q_1, q_2 \in \sigma_h^*$. By the assumption and the definitions, we have $w(\sigma_h^*) \leq w^*(Q) < width(P) \leq L$ and $L \leq d(p, \ell_{q_1 q_2})$ for every $p \in P$. Since $q_1, q_2 \in \sigma_h^*$ and $w(\sigma_h^*) < d(p, \ell_{q_1 q_2})$ for every $p \in P$, it follows that $P \cap \sigma_h^* = \emptyset$. Therefore, σ_t^* should cover P which implies that $w^*(Q) \geq w(\sigma_t^*) \geq \text{width}(P)$. This contradicts our assumption that $w^*(Q) < \text{width}(P)$.

Now, consider the case where $q_1, q_2 \notin \sigma_h^*$. By the assumption and the definitions, $w(\sigma_h^*) \leq w^*(Q) < \text{width}(P) \leq L_2$. Therefore, there exists a point $r \in P$ not contained in σ_h^*. This implies that $q_1, q_2, r \in \sigma_t^*$. It follows that $w(\sigma_t^*) \geq \text{width}(\{q_1, q_2, r\})$. Since $\overline{q_1 q_2}$ is the longest side in the triangle $\triangle q_1 q_2 r$, we have $\text{width}(\{q_1, q_2, r\}) = d(r, \ell_{q_1 q_2})$. As $d(r, \ell_{q_1 q_2}) \geq L$, it follows that $w(\sigma_t^*) \geq L$. Consequently, $w^*(Q) \geq w(\sigma_t^*) \geq L \geq \text{width}(P)$, which contradicts our assumption. □

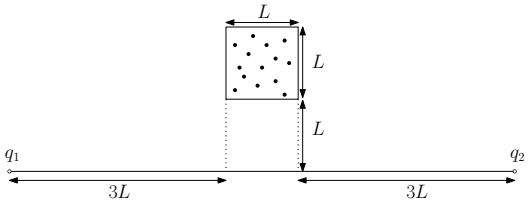


Figure 27: The square of side length L covers the points.

D Ruling Out $w^* = 0$ in Linear Time

General Choose two arbitrary points p and q in S, and find a point $r \in S$ that does not lie on ℓ_{pq}. If no such point exists, then all points of S lie on ℓ_{pq}, which implies $w^* = 0$. Assume such a point r exists. If $w^* = 0$, then at least one slab (of width 0) in an optimal solution must be ℓ_{pq}, ℓ_{pr}, or ℓ_{qr}. We test all three cases. For example, to test whether there exists an optimal pair of slabs where one of them is ℓ_{pq}, we check whether all points of S not lying on ℓ_{pq} lie on a common line. This can be done by choosing two arbitrary points $u, v \in S \setminus \ell_{pq}$ and checking if all points in $S \setminus \ell_{pq}$ lie on ℓ_{uv}. This entire procedure takes linear time.

One fixed orientation By rotating the coordinate system, we assume without loss of generality that the given orientation θ is 0.

Choose two arbitrary points p and q in S, and find a point $r \in S$ that does not lie on ℓ_{pq}. If no such point exists, then all points of S lie on ℓ_{pq}, which implies $w^* = 0$. Assume such a point r exists. If $w^* = 0$, then at least one slab (of width 0) in an optimal solution must be ℓ_{pq}, ℓ_{pr}, or ℓ_{qr}. We test all three cases. We give an example for testing whether there exists an optimal pair of slabs where one of them is ℓ_{pq}. If ℓ_{pq} is horizontal, we check whether all points

of S not lying on ℓ_{pq} lie on a common line. This can be done by choosing two arbitrary points $u, v \in S \setminus \ell_{pq}$ and checking if all points in $S \setminus \ell_{pq}$ lie on ℓ_{uv}.

If ℓ_{pq} is not horizontal, we check whether all points of S not lying on ℓ_{pq} lie on a horizontal line. This can be done by choosing an arbitrary point $u \in S \setminus \ell_{pq}$ and checking if all points in $S \setminus \ell_{pq}$ lie on the horizontal line passing through u. This entire procedure takes linear time.

Two fixed orientations By rotating the coordinate system, we assume without loss of generality that one of the given orientations is horizontal. Thus, the two orientations can be taken as 0 and θ, where $0 \leq \theta < \pi$. Let (σ_h^*, σ_t^*) be an optimal pair of slabs where σ_h^* is horizontal and σ_t^* has orientation θ.

Choose an arbitrary point $p \in S$, and then find another point $q \in S$ with a different y-coordinate from p. If no such point exists, then all points of S lie on a horizontal line, which implies $w^* = 0$. If such q exists, then for $w^* = 0$ it must be the case that p or q (or possibly both) lies on σ_t^*. Let ℓ_p be the line of orientation θ passing through p, and define ℓ_q analogously for q. If $w^* = 0$, then σ_t^* must be either ℓ_p or ℓ_q. We test both cases by checking whether all points of S not lying on ℓ_p (resp., ℓ_q) lie on a common horizontal line. The procedure above takes linear time in total.

Parallel Choose two arbitrary points p and q in S, and find a point $r \in S$ that does not lie on ℓ_{pq}. If no such point exists, then all points of S lie on ℓ_{pq}, which implies $w^* = 0$. Assume such a point r exists. If $w^* = 0$, then at least one slab (of width 0) in an optimal solution must be ℓ_{pq}, ℓ_{pr}, or ℓ_{qr}. We test all three cases. For example, to test whether there exists an optimal pair of slabs where one of them is ℓ_{pq}, we check whether all points of S not lying on ℓ_{pq} lie on the line parallel to ℓ_{pq} that passes through r. This entire procedure takes linear time.